# EIN 4343 Inventory and Supply Chain Systems

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

# INVENTORY SYSTEMS: MODELING AND ANALYSIS

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# A blueprint going forward

- Modeling of demand
  - Deterministic vs. stochastic demand process
- Planning horizon length
  - Infinite vs. finite planning horizon

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# INVENTORY SYSTEMS: EOQ: DETERMINISTIC DEMAND INFINITE HORIZON

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# Economic Order Quantity (EOQ) Model

- Simplest and most fundamental of all inventory models

- Trade-off between the fixed order cost and variable inventory holding cost

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# EOQ: Notation

$\lambda$: demand rate (units/unit time)

$K$: fixed order cost ($/order)

$c$: unit variable purchase cost ($/unit)

$h$: unit inventory holding cost ($/unit/unit time)

**Parameters**

$q$: order quantity (units);

$T$: length of the inventory cycle (unit time);

**Decision variables**

$C(q)$: average total cost per unit time ($/unit time).

**Objective function**

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# EOQ: Modeling assumptions

1. The planning horizon is very long (i.e., infinite).
2. The inventory level is reviewed on a continuous basis.
3. The demand arrives at a known, constant, and continuous rate over time, and the parameter does not change throughout the planning horizon.
4. The entire order quantity can be delivered at the same time.
5. The order quantity need not be an integral number of units, and there are no minimum or maximum restrictions on the order quantity.

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# EOQ: Modeling assumptions

6. Although the item of interest can be an item in a larger inventory system, it is treated entirely independently of other items (i.e., single item).
7. The cost parameters do not change over time. In particular, the inflation rate is low.
8. The unit variable purchase cost is constant and does not depend on order quantity.
9. The order delivery lead time is zero, i.e., instantaneous replenishment.
10. No shortages are allowed, i.e., all the demand must be satisfied.

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# EOQ Model: Why?

Let's do a quick thought experiment.

- Suppose that your fixed order cost of purchasing groceries (say Uber ride to Trader Joe's is $10) and you want to buy a banana. Suppose that keeping a banana in inventory for a day costs you $0.01.
  - Buy 1 banana. Eat it. Inventory holding cost is 0.
  - Buy 7 bananas. Eat one per day. Weekly inventory holding cost is $0.21.

$6(0.01)+5(0.01)+\ldots+1(0.01)=\$0.21$

- If you buy 1 at a time, you go to the store every day.
  - You are going to the store every day!
- If you buy 7 at a time, you go the store once a week.
  - You have too many bananas sitting on your counter!
- There is a trade-off between the fixed order cost and the inventory holding cost.

There is a sweet spot. That sweet spot lies where the average total cost per unit time is minimized.

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# EOQ Model: Analytical approach

In order to minimize the average cost per unit time:

1. Determine the length of an inventory cycle
2. Calculate the total cost per inventory cycle
3. Divide the total cost per inventory cycle by the total length of the inventory cycle to find the average total cost per unit time
4. Use continuous optimization techniques to determine the order quantity that minimizes the average total cost per unit time.

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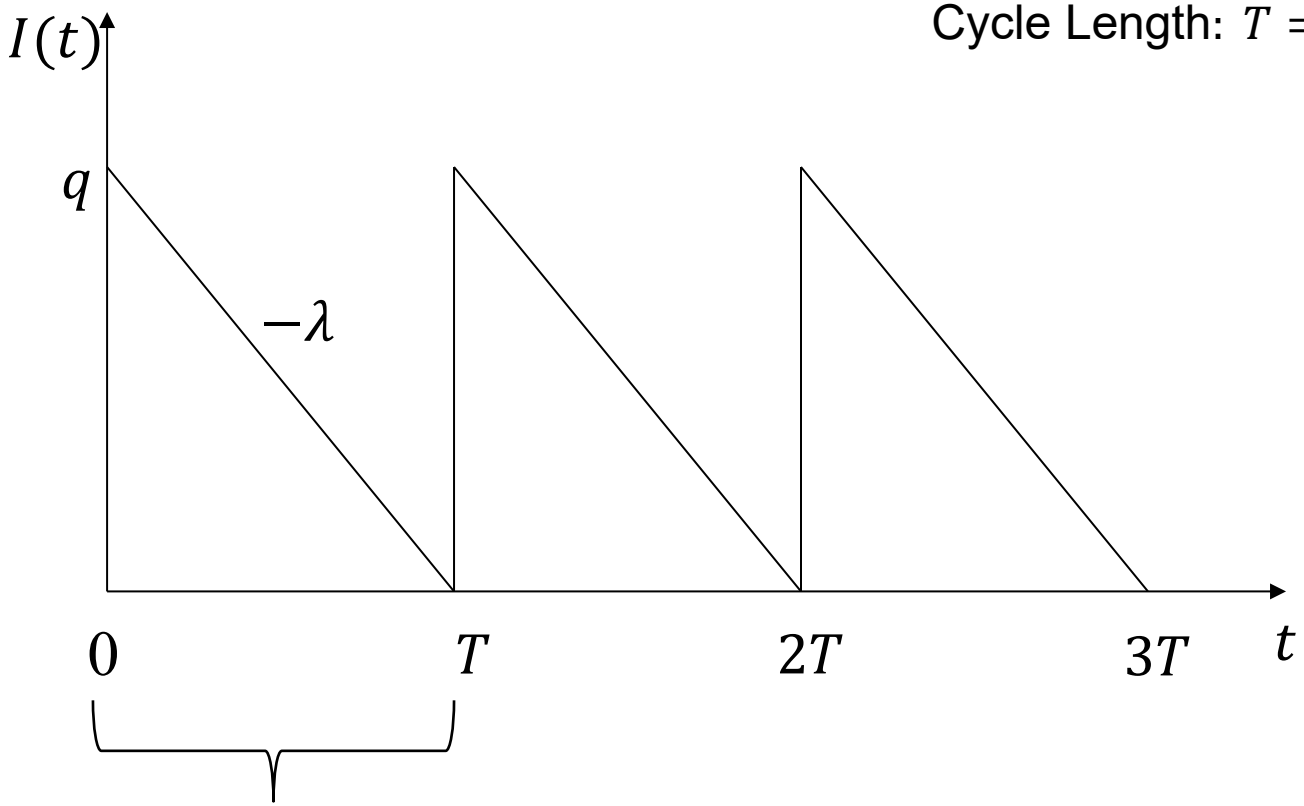
# EOQ Model: Inventory cycle

Cycle Length: $T = \frac{q}{\lambda}$




An inventory cycle

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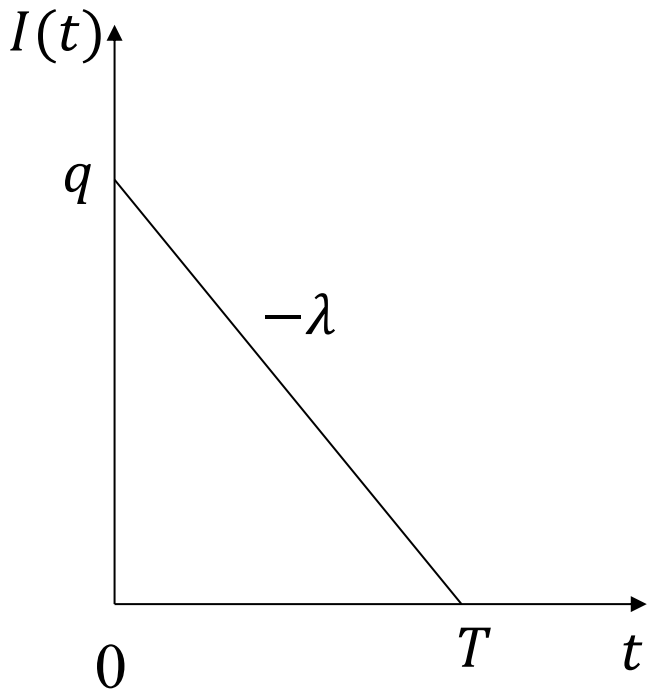
# EQO Model: Total cost per cycle




$$\text{Purchasing Cost } = cq$$

$$\text{Ordering Cost } = K$$

$$\text{Inventory Holding Cost} = h\frac{qT}{2} = \frac{hq^2}{2\lambda}$$

$$\text{Total Cost per Cycle } = cq + K + \frac{hq^2}{2\lambda}$$

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# EOQ Model: Average total cost per unit time

$$\text{Average Total Cost/Unit time} = \frac{\text{Total Cost per Cycle}}{\text{Cycle Length}} = C(q)$$

$$C(q) = \frac{cq + K + \frac{hq^2}{2\lambda}}{\frac{q}{\lambda}}$$

$$C(q) = \boxed{c\lambda} + \boxed{\frac{K\lambda}{q}} + \boxed{\frac{hq}{2}}$$

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# EOQ Model: Optimization

- Optimization problem of interest

$$\min C(q)$$

$$\text{subject to}$$

$$q > 0$$

- Take the first derivative, set it equal to zero. Take the second derivative, check convexity.

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# EOQ Model: Optimization

$$C(q) = c\lambda + \frac{K\lambda}{q} + \frac{hq}{2}$$

$$\frac{dC}{dq} = -\frac{K\lambda}{q^2} + \frac{h}{2}$$

$$\frac{dC}{dq} = 0 \implies q^* = \sqrt{\frac{2K\lambda}{h}}$$

$$\frac{d^2C}{dq^2} = \frac{2K\lambda}{q^3} > 0$$

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# EOQ Model: An observation based on the first order condition

- Recall the first order condition

$$\frac{dC}{dq} = -\frac{K\lambda}{q^2} + \frac{h}{2} = 0$$

- Manipulate the first order condition algebraically

$$\frac{K\lambda}{q} = \frac{hq}{2}$$

- What do we have?

Annual ordering cost = Annual inv. hold. cost

- What do you think?

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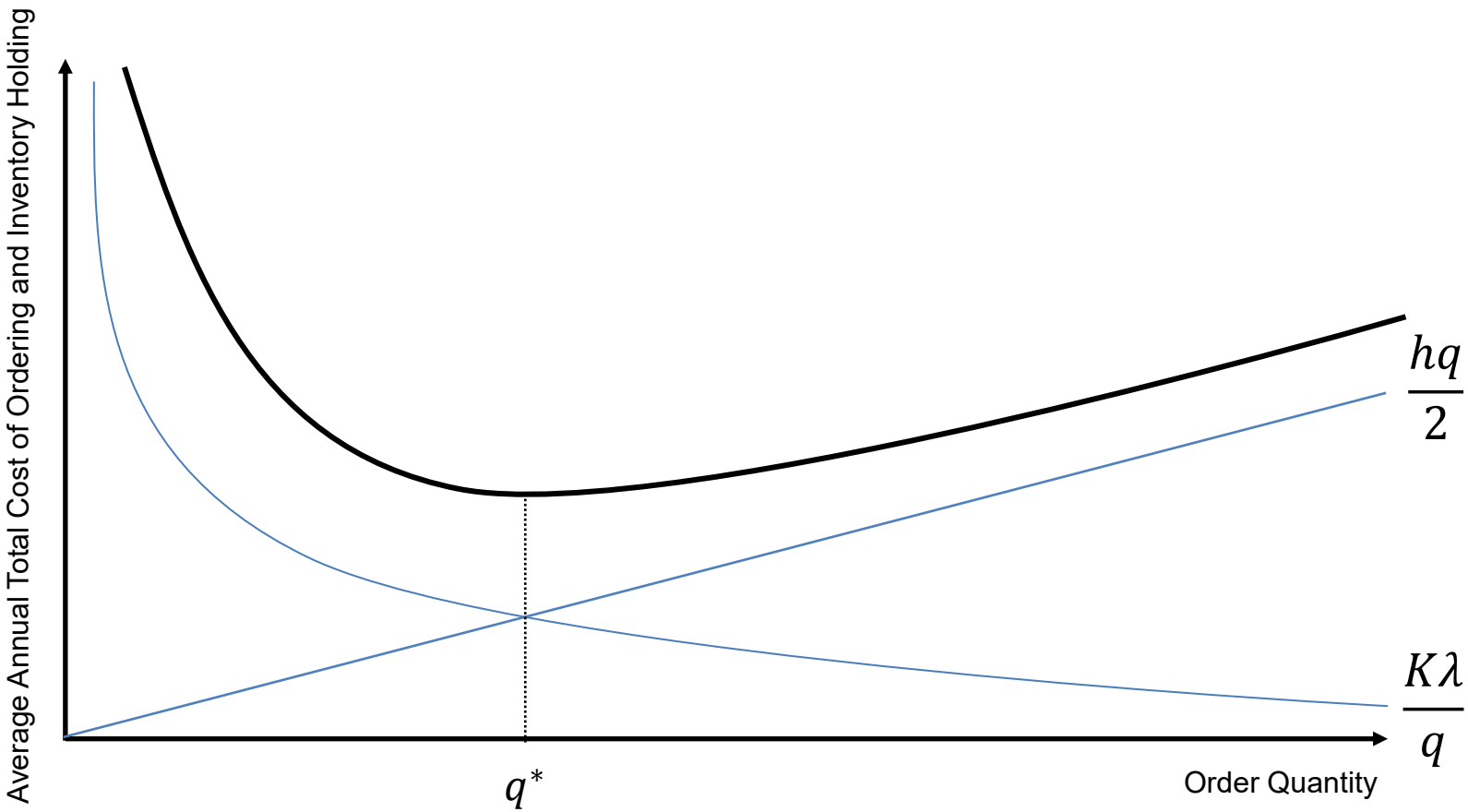
# EOQ Model: Insight

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# More on the objective function

- Average annual ordering cost

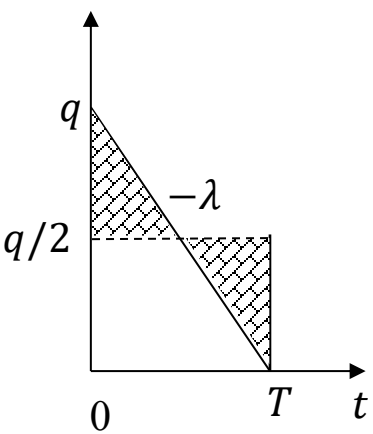
(order cost/order) (no. of orders per year)

$$K\frac{\lambda}{q}$$

- Average annual inventory holding cost

(inventory holding cost/unit/year) (average level of inventory throughout the year)

$$h\frac{q}{2}$$




Graphical insight: Cut the top half of the triangle representing the total inventory held per cycle, rotate 180 degrees, and slide down.

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# EOQ: Example (1 of 3)

Rice is sold at a fairly steady rate of 40 lbs per week at Krishna lunch on UF campus. Rice costs the organizers \$0.2/lb. It costs \$20 to place and receive an order. It costs Krishna \$0.05 to keep a pound of rice in inventory for a year. Assume that Krishna lunch is organized for 46 weeks out of the year.

Annual demand rate: $\lambda = (46)(40) = 1{,}840 \text{ lbs/year}$

Fixed order cost: $K = \$20/\text{order}$

Unit purchase cost: $\text{c} = \$0.20/\text{lb}$

Unit inventory holding cost: $h = \$0.05/\text{lb/year}$

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# EOQ: Example (2 of 3)

a. What is the optimal order quantity for rice for Krishna?

$$q^* = \sqrt{\frac{2K\lambda}{h}} = \sqrt{\frac{(2)(20)(1{,}840)}{0.05}} = \sqrt{1{,}472{,}000} = 1{,}213.26 \text{ lbs}$$

Do NOT round optimal order quantity down or up to an integer value. Use two significant digits after the decimal point unless you are explicitly asked to.

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# EOQ: Example (3 of 3)

b. What is the total average annual ordering and inventory holding cost for rice?

$$C(q) = \frac{K\lambda}{q} + \frac{hq}{2} \quad = \frac{(20)(1{,}840)}{1{,}213.26} + \frac{(0.05)(1{,}213.26)}{2}$$

$$= 30.33 + 30.33$$

$$= \$60.66/\text{year}$$

Are you surprised that the average annual ordering and inventory holding cost terms are equal? No, you should not be. Recall Slide 15.

Note that $C(q)$ has another term, annual purchasing cost, $c\lambda$, which is not included here.

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# INVENTORY HOLDING COST AND INVENTORY CARRYING RATE

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# Definitions

$h$ is the annual inventory holding cost for the item in terms of \$/unit/year and is given by

$$h = ic$$

where

$i$ is the annual inventory carrying rate for the system; and

$c$ is the unit cost of an item.

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# Annual inventory carrying rate

Inventory holding cost

The opportunity cost associated with the investment in inventory

Expenses that a company incurs for holding inventory

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# Annual inventory carrying rate

## How to calculate inventory holding cost expenses?

- Typically, a firm can calculate the total expenses incurred for inventory

- The firm can determine a rate known as the *inventory carrying rate*, denoted by $i$, and spread the related costs over the existing inventory based on the value of the inventory

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# Example: Annual inv. carrying rate

A manufacturing company supplies spare parts for industrial equipment and keeps an inventory mostly made up of machine parts. The inventory consists of 5,000 SKUs valued at $525,000 (valued at full cost). The company rents the warehouse space for $30,000 per year. The main operating costs (in $) per year in the warehouse are estimated to be as follows:

| Cost item | Cost ($/year) |
|---|---:|
| Municipal and other taxes | 5,732.67 |
| Insurance on building and its contents | 3,200.00 |
| Heating and air conditioning | 12,500.00 |
| Electricity and water | 3,200.00 |
| Labor (supervisor, attendants, etc.) | 69.000.00 |
| Pilferage | 5,000.00 |
| Obsolescence | 5,000.00 |

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# Example: Annual inv. carrying rate

Manufacturing overhead is charged against the material cost. Manufacturing overhead rate used by the company is 60% and it is estimated that on average each SKU involved one dollar of labor cost per dollar of material cost. Recommend a value for inventory carrying rate $i$.

*Hint: If the inventory is valued at full cost, then it means that direct (material and labor) as well as indirect (manufacturing overhead) costs are taken into account. In this case, in order to find the money invested in inventory, you need to remove the manufacturing overhead cost.*

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# Solution

An SKU valued at full cost has $1 of labor, $1 of material and $0.60 of overhead.

$$\left(\frac{1+1}{1+1+0.6}\right)525{,}000 = \left(\frac{2}{2.6}\right)525{,}000 = 403{,}846.15$$

Total expenses

$$5{,}732.67 + 3{,}200 + \cdots + 5{,}000 + 30{,}000 = 133{,}632.67$$

Inventory carrying rate due to out-of-pocket expenses

$$\frac{133{,}632.67}{403{,}846.15} = 0.3309 \equiv 33.09\%$$

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# Question

- Where do 2.6 and 2 come from?

- Recall that
  - $1 labor cost, $1 material cost, and $0.60 manufacturing overhead charged against $1 material cost.

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# Trader Joe’s Feta Cheese

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

# INVENTORY SYSTEMS: DETERMINISTIC DEMAND INFINITE HORIZON EXTENSIONS

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# EOQ: Modeling assumptions

1. The inventory level is reviewed continuously.
2. The planning horizon is very long (i.e., infinite), and the demand parameter will continue to be at the same value throughout this planning horizon.
3. The demand arrives at a known, constant, and continuous rate over time.
4. The entire order quantity can be delivered at the same time.
5. The order quantity need not be an integral number of units, and there are no minimum or maximum restrictions on the order quantity.
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7. The cost parameters do not change over time. In particular, the inflation rate is low.
8. The unit variable purchase cost is constant and does not depend on order quantity.
9. **The order delivery lead time is zero, i.e., instantaneous replenishment.**
10. No shortages are allowed, i.e., all the demand must be satisfied.

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# Constant lead time

- *Lead time* is the time between when an order is placed and when it is received

- Let $\tau$ denote the constant lead time.
  - That is, when an order is placed, it is received after $\tau$ time units.

- How does this change the optimal order quantity?

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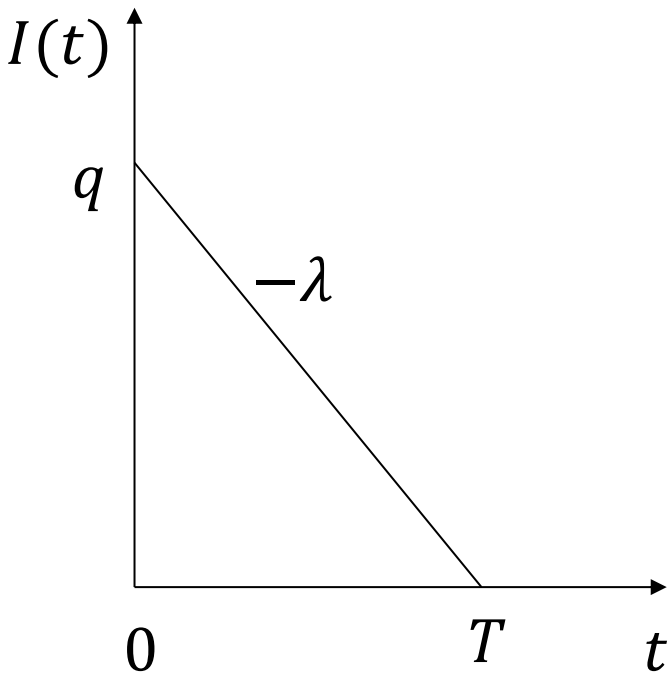
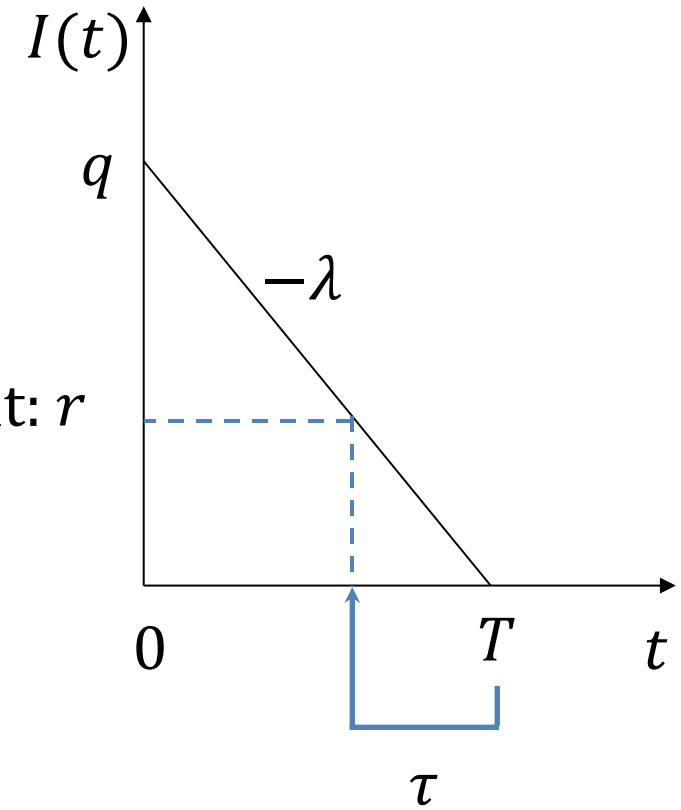
# EOQ: Constant lead time




Reorder Point: $r$




Instantaneous replenishment → Known lead time, $\tau$

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# Demand during lead time

- What is $\lambda\tau$?
  - The multiplication of the annual demand rate parameter with the constant lead time parameter
  - Demand during lead time

- Two cases to consider
  - What if $\lambda\tau < q$?
  - What if $\lambda\tau > q$?

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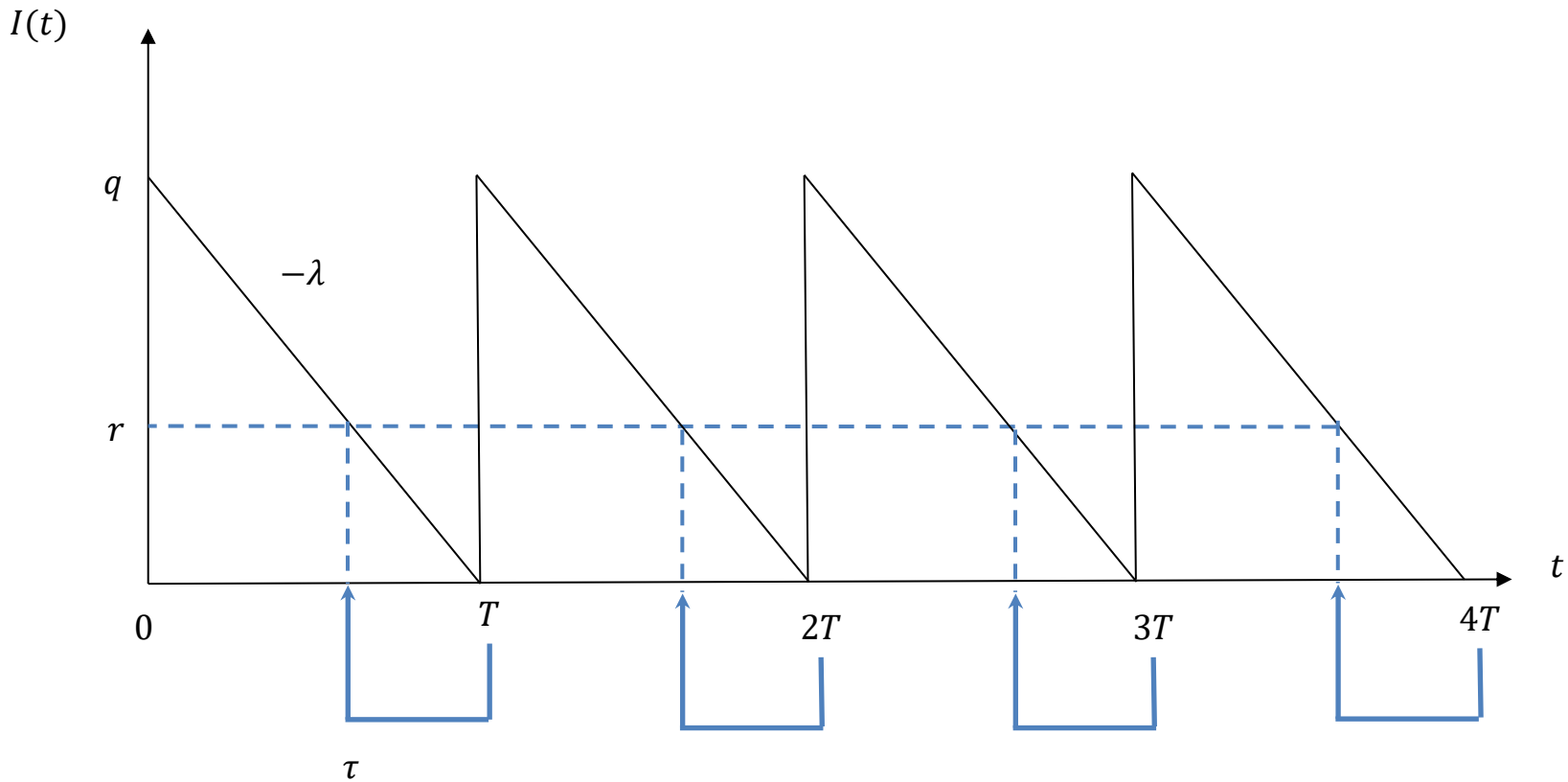
# What if: $\tau < T$ or $\lambda\tau < q$




Case 1: $\lambda\tau < q$

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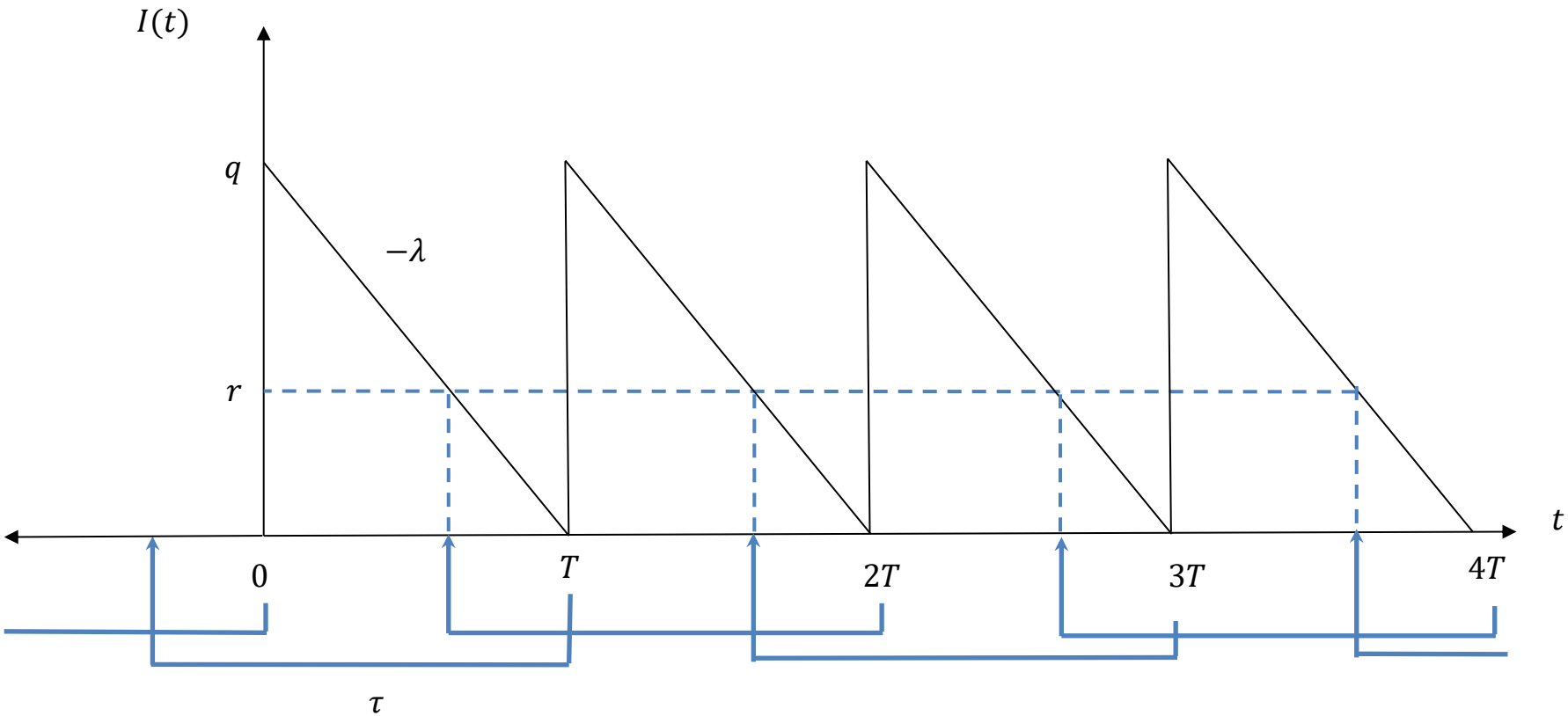
# What if: $\tau > T$ or $\lambda\tau > q$




Case 2: $\lambda\tau > q$

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# EOQ Model: Constant lead time

- Recall that

  $\lambda\tau$: demand during lead time (units) and

  $q$: order quantity (units).

- Reorder level, $r$, is given by

$$r = \mathrm{mod}(\lambda\tau, q)$$

- Recall that $\mathrm{mod}(x, y)$ is given by dividing $x$ by $y$ and returning the remainder.

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# Example

- Suppose that the optimal order quantity for an item is 400 units, and the annual demand rate is 2,400 units/year.

- Based on a 50-week work year, specify the reorder level if the lead time is
  a) 3 weeks; or
  b) 12 weeks.

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# Solution

a) Lead time is 3 weeks

$$\lambda\tau = 2{,}400\frac{3}{50} = 144 \text{ units}$$

$$r = \text{mod}(144{,}400) = 144 \text{ units}$$

b) Lead time is 12 weeks

$$\lambda\tau = 2{,}400\frac{12}{50} = 576 \text{ units}$$

$$r = \text{mod}(576{,}400) = 176 \text{ units}$$

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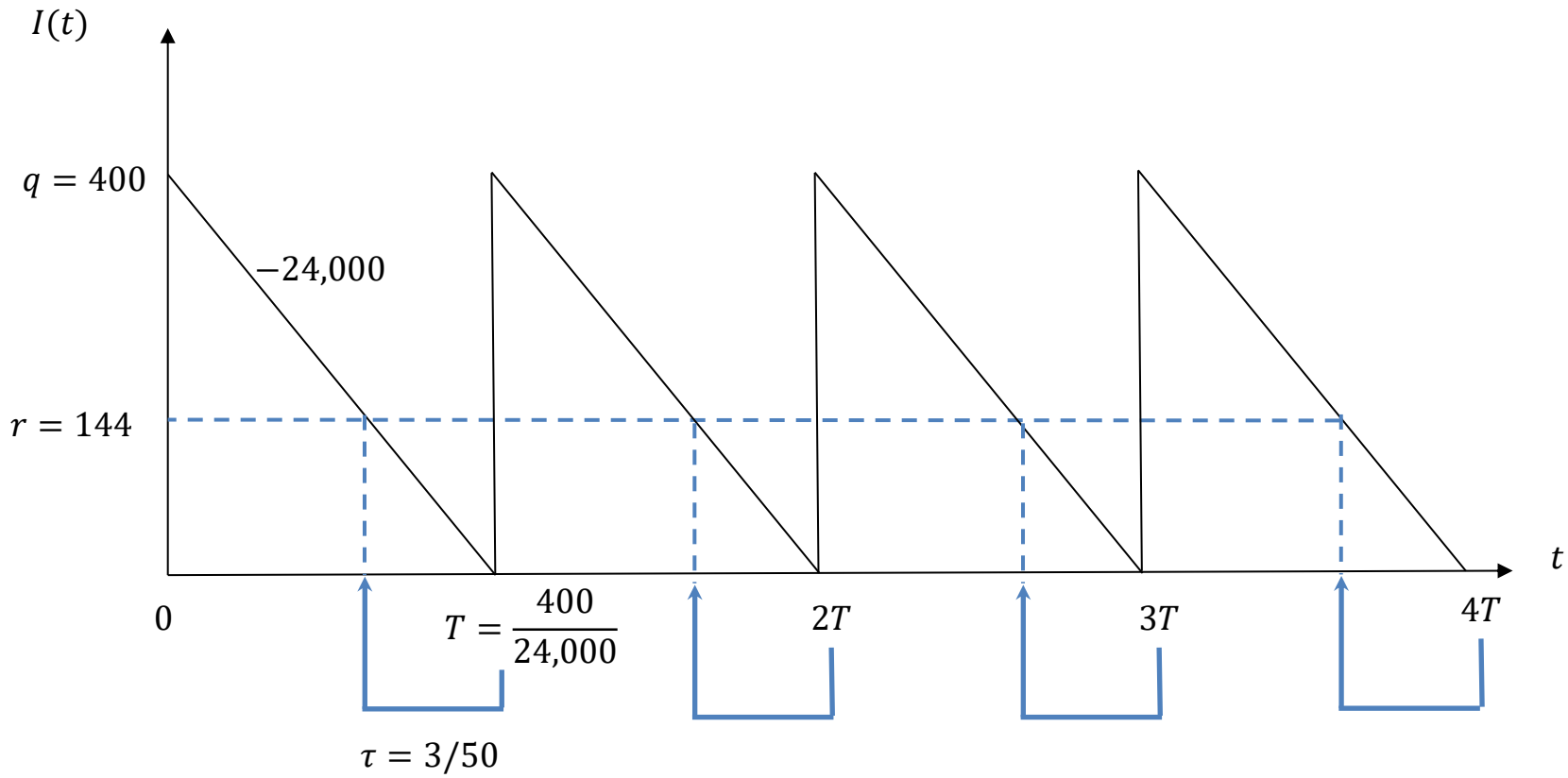
# Solution: Demonstration $\tau = 3$ weeks




Case 1: $\lambda\tau < q$

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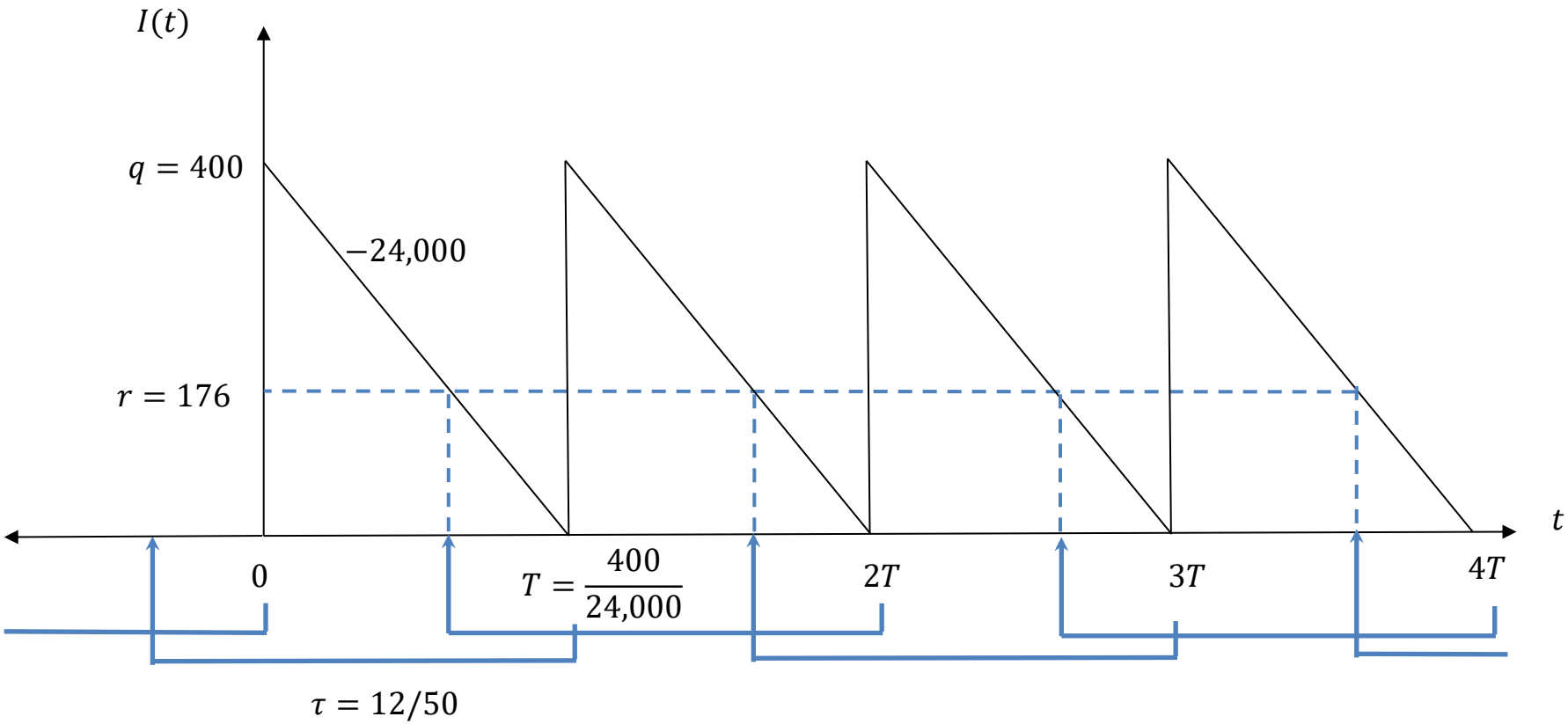
# Solution: Demonstration $\tau = 12$ weeks




Case 2: $\lambda\tau > q$

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# EOQ: Modeling assumptions

1. The inventory level is reviewed continuously.
2. The planning horizon is very long (i.e., infinite), and the demand parameter will continue to be at the same value throughout this planning horizon.
3. The demand arrives at a known, constant, and continuous rate over time.
4. **The entire order quantity can be delivered at the same time.**
5. The order quantity need not be an integral number of units, and there are no minimum or maximum restrictions on the order quantity.
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10. No shortages are allowed, i.e., all the demand must be satisfied.

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# EOQ Model: Finite replenishment rate

- Consider a production setting.

- Let $P$ denote the constant and continuous production rate.
  - When an order of size $q$ is placed, the delivery of items at the continuous rate of $P$ starts immediately and continues until a total of $q$ units are delivered.

- How does this change the optimal order quantity?
  - Economic Production Quantity (EPQ)

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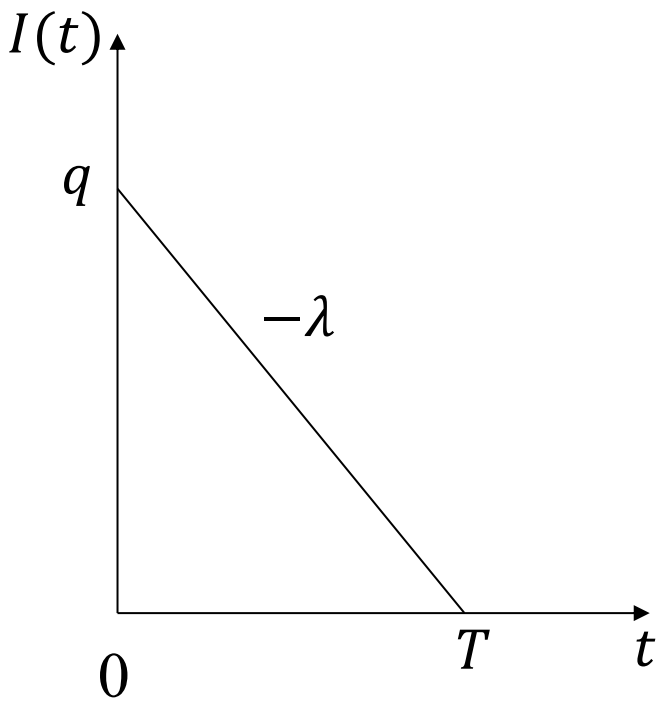
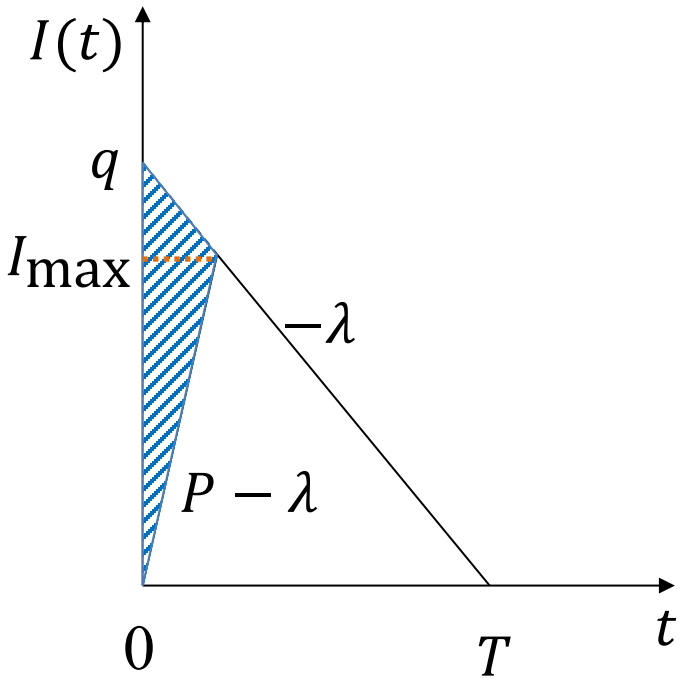
# EPQ Model: Cycle length




Infinite replenishment rate → $P$ : Production rate → Finite Replenishment rate

$$\boxed{P > \lambda}$$

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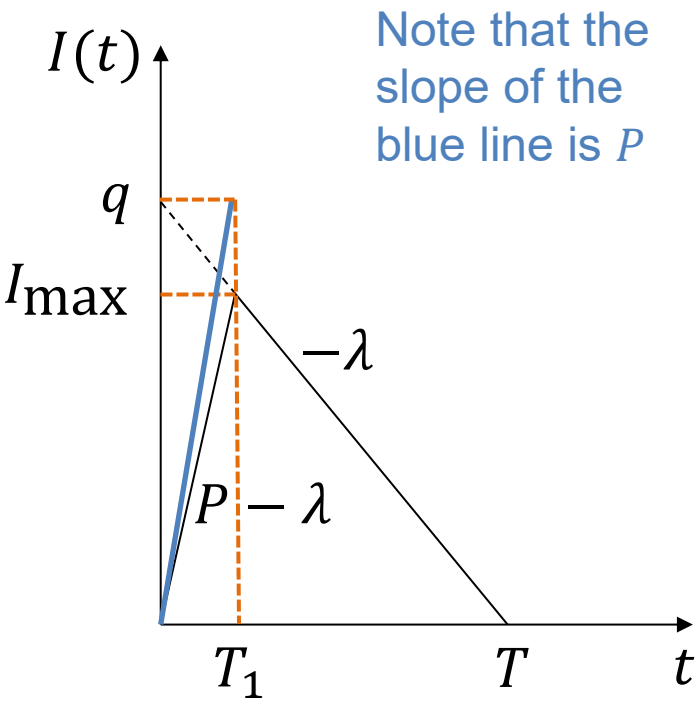
# EPQ Model: Maximum inventory level

Note that $T_1$ is <u>the time it takes to produce $q$ units at a rate of $P$</u>. We have

$$T_1 = \frac{q}{P}$$

Note also that $T_1$ is <u>the time it takes to add $I_{\max}$ units in inventory at a rate of $P - \lambda$</u>. We have

$$T_1 = \frac{I_{\max}}{P - \lambda}$$




Hence, we have

$$\frac{q}{P} = \frac{I_{\max}}{P - \lambda} \Rightarrow I_{\max} = q\left(1 - \frac{\lambda}{P}\right)$$

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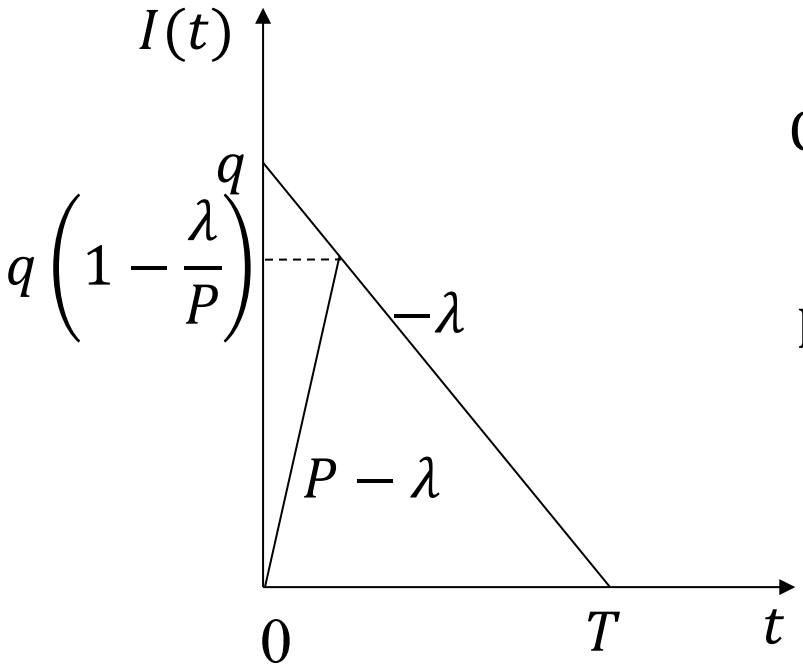
# EPQ Model: Total cost in an inventory cycle




$$\text{Purchasing Cost } = cq$$

$$\text{Ordering Cost} = K$$

$$\text{Inventory Holding Cost} = h\frac{q\left(1-\frac{\lambda}{P}\right)T}{2}$$

$$cq + K + h\left(1-\frac{\lambda}{P}\right)\frac{qT}{2}$$

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# EPQ Model: Average total cost per unit time

$$\text{Average Total Cost/Unit time} = \frac{\text{Total Cost per Cycle}}{\text{Cycle Length}} = C(q)$$

$$C(q) = \frac{cq + K + h\left(1 - \frac{\lambda}{P}\right)\frac{qT}{2}}{\frac{q}{\lambda}}$$

$$C(q) = c\lambda + \frac{K\lambda}{q} + \frac{h\left(1 - \frac{\lambda}{P}\right)q}{2}$$

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# EPQ Model: Optimization

$$C(q) = c\lambda + \frac{K\lambda}{q} + \frac{h\left(1 - \frac{\lambda}{P}\right)q}{2}$$

$$\frac{dC}{dq} = -\frac{K\lambda}{q^2} + \frac{h\left(1 - \frac{\lambda}{P}\right)}{2}$$

$$\frac{dC}{dq} = 0 \implies q^* = \sqrt{\frac{2K\lambda}{h\left(1 - \frac{\lambda}{P}\right)}}$$

$$\frac{d^2C}{dq^2} = \frac{2K\lambda}{q^3} > 0$$

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# Example

Suppose that the item costs $30/unit to produce and the fixed cost of production setup is $500/order. For this item suppose that the annual demand rate is 24,000 units/year whereas the annual production rate is 50,000 units/year. Suppose that an annual inventory carrying rate of 0.25 is in effect.

a. What is the optimal production quantity for this item?
b. What is the average level of inventory in the system when the optimal production quantity you identified in part (a) were to be used?
c. What is the average annual total (production, production setup and inventory carrying) cost associated with this item?
d. Compare the total cost obtained in part (c) under a finite production rate of 50,000 units/year to the average annual total cost under an infinite rate.

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# Solution: Parameters

| | |
|---|---|
| Fixed ordering cost: | $K = \$500/\text{setup}$ |
| Unit production cost: | $c = \$30/\text{unit}$ |
| Ann. inventory carrying rate: | $i = 0.25/\text{year}$ |
| Inventory holding cost: | $h = ic = (0.25)(30) = \$7.5\ \text{unit/year}$ |
| Annual demand rate: | $\lambda = 24{,}000\ \text{units/year}$ |
| Annual production rate: | $P = 50{,}000\ \text{units/year}$ |

Is the production rate greater than demand rate? That is, is $\mathrm{P} > \lambda$ true?

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# Solution: Part (a)

$$q^* = \sqrt{\frac{2K\lambda}{h\left(1-\frac{\lambda}{P}\right)}} = \sqrt{\frac{2(500)(24{,}000)}{(7.5)\left(1-\frac{24{,}000}{50{,}000}\right)}} = \sqrt{\frac{2(500)(24{,}000)}{(7.5)(0.52)}}$$

$$= 2{,}480.69 \text{ units}$$

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# Solution: Part (b)

- First, we begin by identifying the maximum inventory level

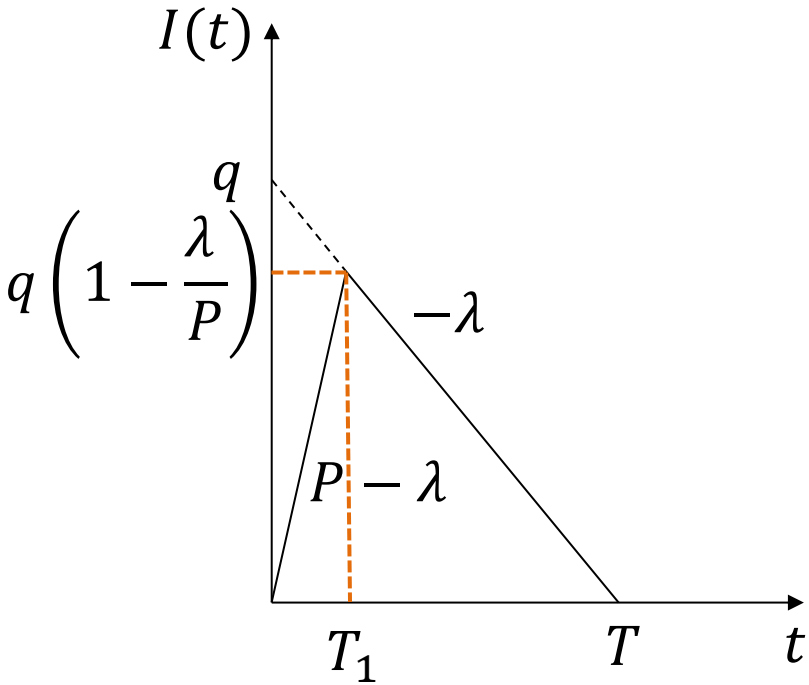
$$I_{\max} = q\left(1 - \frac{\lambda}{P}\right)$$

$$= 2{,}480.69\left(1 - \frac{24{,}000}{50{,}000}\right)$$

$$= 1{,}289.96 \text{ units}$$

- Next, we identify the average inventory level which is half of the maximum inventory level. Hence, we have

$$\frac{1{,}289.96}{2} = 644.98 \text{ units}$$

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# Solution: Part (c)

$$C(q) = c\lambda + \frac{K\lambda}{q} + \frac{h\left(1 - \frac{\lambda}{P}\right)q}{2}$$

$$= (30)(24{,}000) + \frac{(500)(24{,}000)}{2{,}480.69} + \frac{7.5\left(1 - \frac{24{,}000}{50{,}000}\right)2{,}480.69}{2}$$

$$= 720{,}000 + 4{,}837.36 + 4{,}837.35$$

$$= \$729{,}674.71/\text{year}$$

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# Solution: Part (d)

**Finite production rate (EPQ)**

$$q_{EPQ}^* = \sqrt{\frac{2K\lambda}{h\left(1-\frac{\lambda}{P}\right)}}$$

$$= 2{,}480.69 \text{ units}$$

$$C(q_{EPQ}^*) = c\lambda + \frac{K\lambda}{q_{EPQ}^*} + \frac{h\left(1-\frac{\lambda}{P}\right)q_{EPQ}^*}{2}$$

$$= 720{,}000 + 4{,}837.36 + 4{,}837.35$$

$$C(q_{EPQ}^*) = \$729{,}674.71/\text{year}$$

**Infinite production rate (EOQ)**

$$q_{EOQ}^* = \sqrt{\frac{2K\lambda}{h}}$$

$$= 1{,}788.85 \text{ units}$$

$$C(q_{EOQ}^*) = c\lambda + \frac{K\lambda}{q_{EOQ}^*} + \frac{hq_{EOQ}^*}{2}$$

$$= 720{,}000 + 6{,}708.22 + 6708.19$$

$$C(q_{EOQ}^*) = \$733{,}416.41/\text{year}$$

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# EOQ: Modeling assumptions

1. The inventory level is reviewed continuously.
2. The planning horizon is very long (i.e., infinite), and the demand parameter will continue to be at the same value throughout this planning horizon.
3. The demand arrives at a known, constant, and continuous rate over time.
4. The entire order quantity can be delivered at the same time.
5. The order quantity need not be an integral number of units, and **there are no minimum or maximum restrictions on the order quantity.**
6. Although the item of interest can be an item in a larger inventory system, it is treated entirely independently of other items (i.e., a single-item system).
7. The cost parameters do not change over time. In particular, the inflation rate is low.
8. The unit variable purchase cost is constant and does not depend on order quantity.
9. The order delivery lead time is zero, i.e., instantaneous replenishment.
10. No shortages are allowed, i.e., all the demand must be satisfied.

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# Limitation on order size

- So far, we have assumed that the optimal lot size is feasible.

- In practice, there might be a limitation on order quantity
  - Imposed by the seller, i.e., no order larger/smaller than $x$ units
  - Imposed by the carrier, i.e., no delivery larger/smaller than $x$ units

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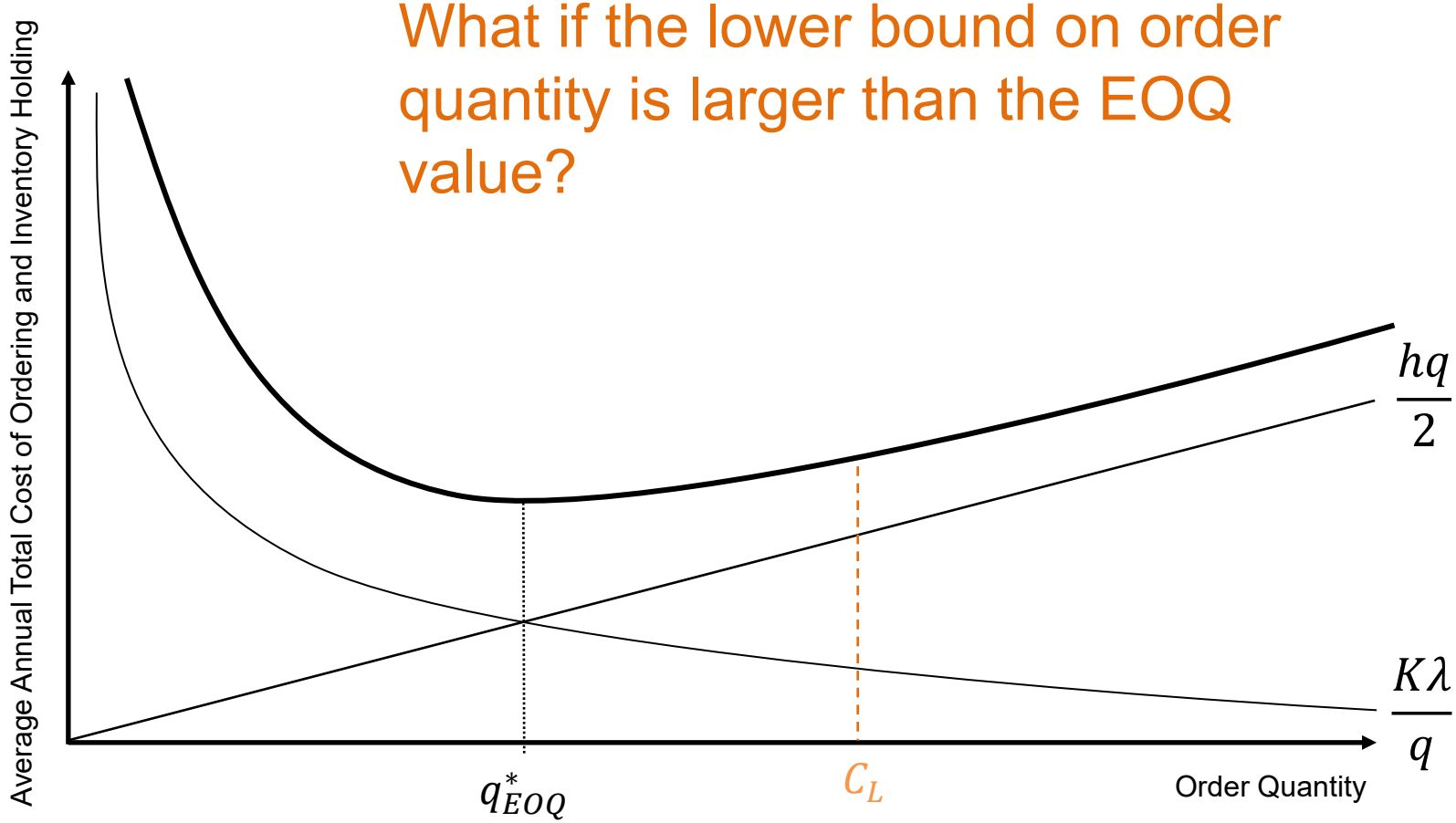
# Limitation on order size: Insight

What if the lower bound on order quantity is larger than the EOQ value?

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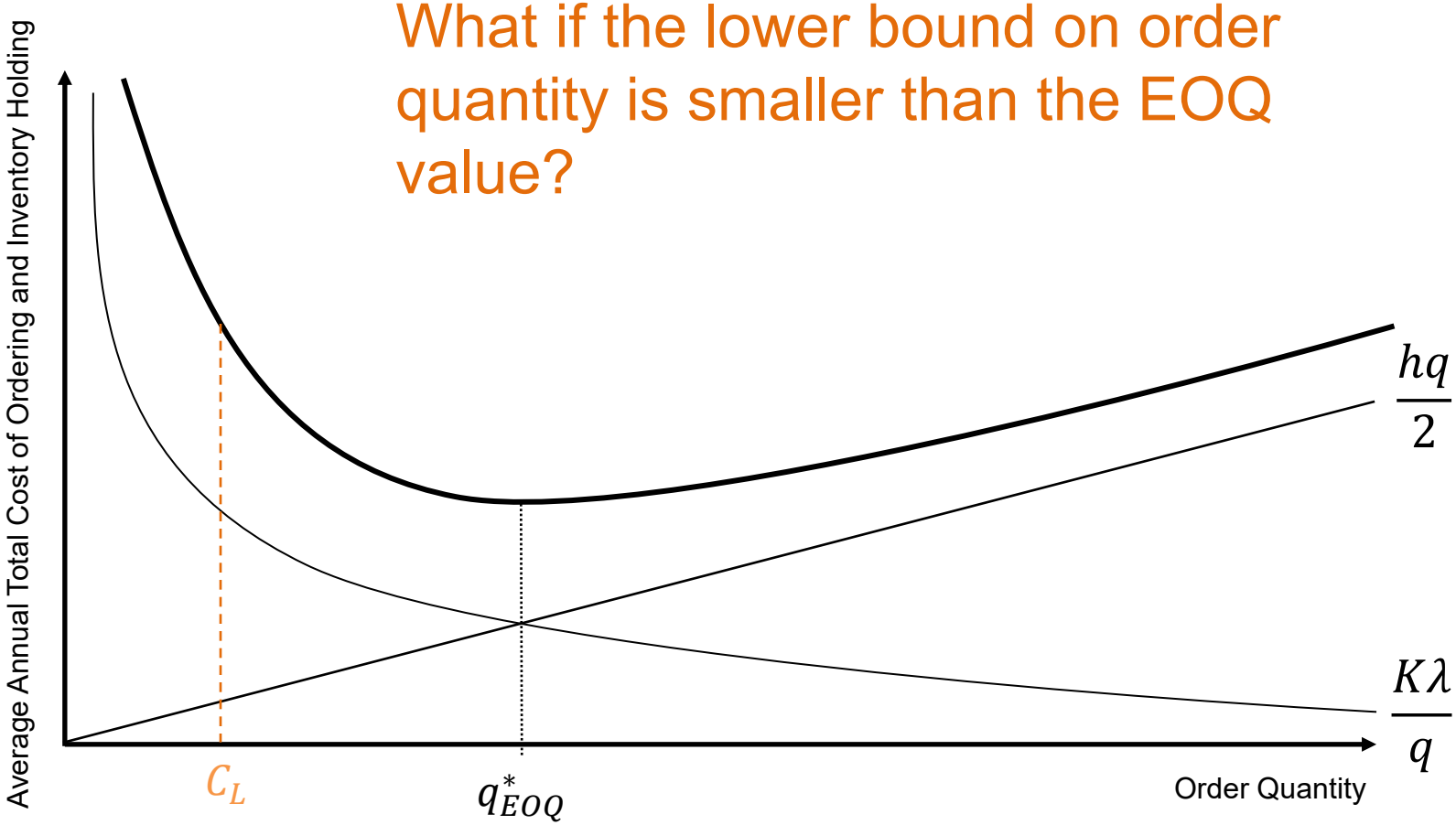
# Limitation on order size: Insight

What if the lower bound on order quantity is smaller than the EOQ value?

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# A lower bound on order quantity

- Let $C_L$ denote the lower bound on order quantity.
- In this case, the optimal lot size is given by

$$q^* = \max\{q^*_{EOQ}, C_L\}$$

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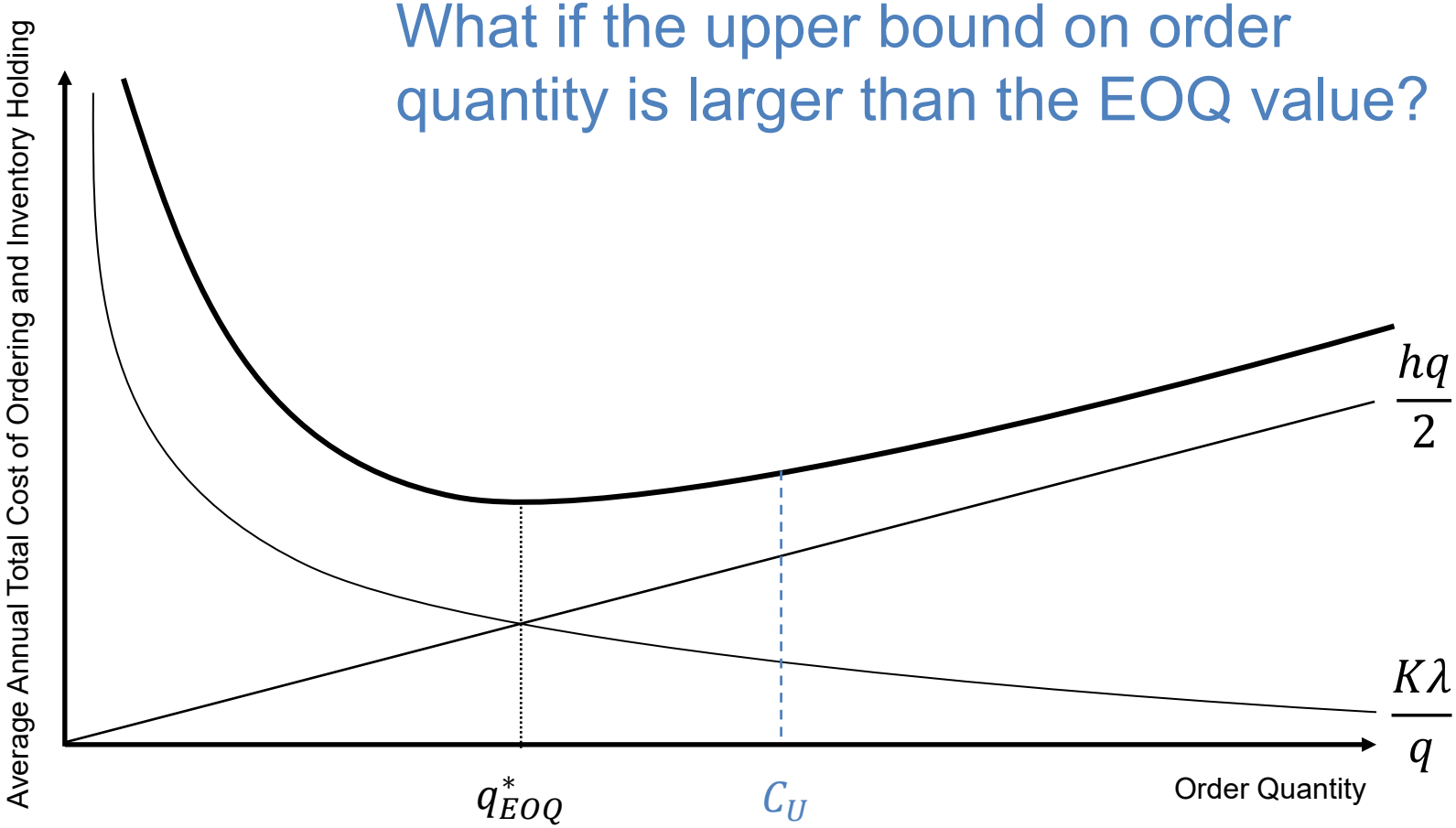
# Limitation on order size: Insight

What if the upper bound on order quantity is larger than the EOQ value?

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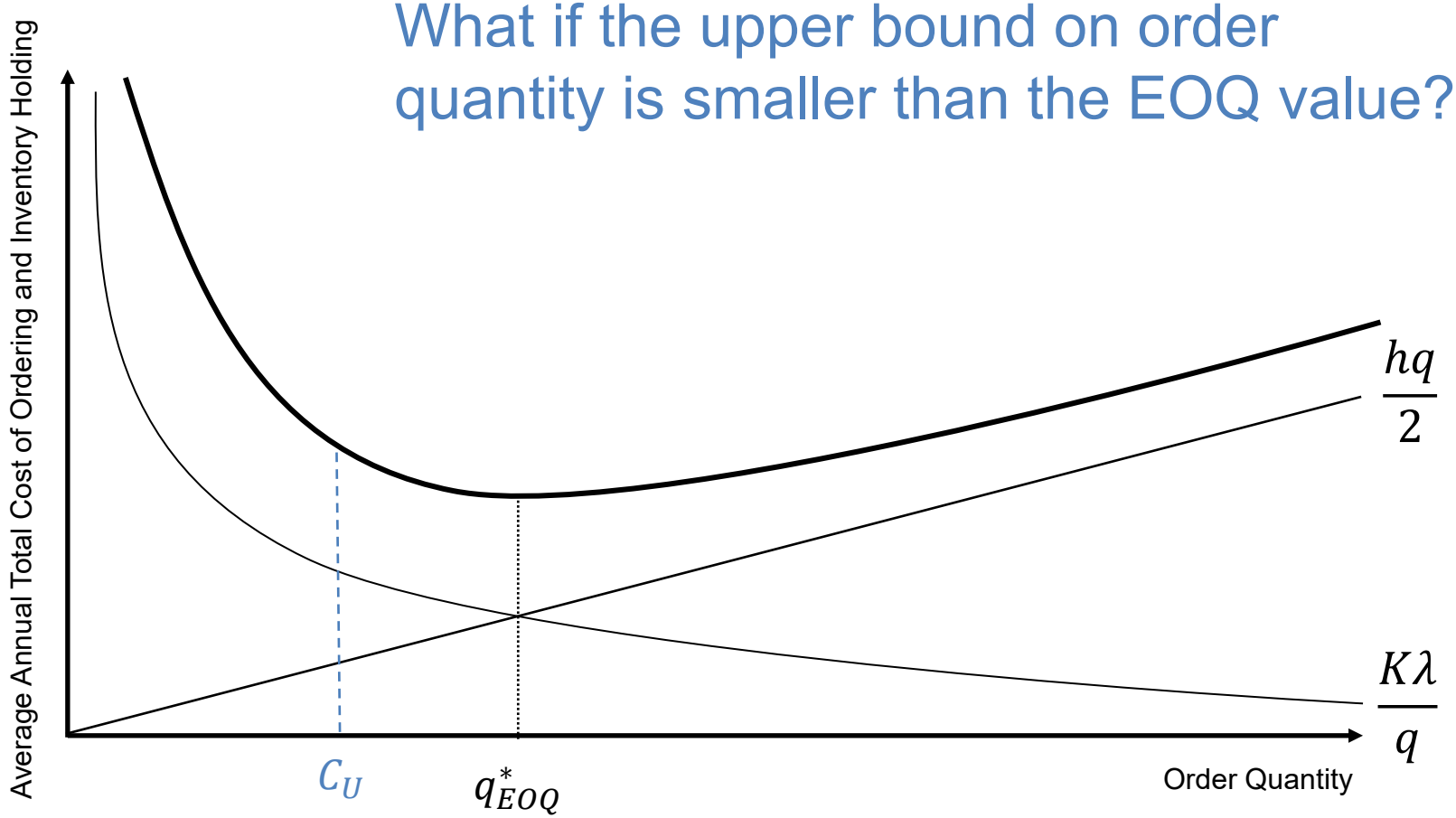
# Limitation on order size: Insight

What if the upper bound on order quantity is smaller than the EOQ value?

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# An upper bound on order quantity

- Let $C_U$ denote the upper bound on order quantity.

- In this case, the optimal lot size is given by

$$q^* = \min\{q^*_{EOQ}, C_U\}$$

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# My two cents

Do not try to memorize and rely on formulas but understand the principles behind it based on the fundamental concept you know

- Sketch the cost function (as the side view of a bowl); a convex function.
- Identify the lowest point of the cost function where EOQ would be; the minimizer point of the convex function.
- Draw a line and see where you can be; add the constraint and identify the feasible region
- See if the optimal solution is at the bound (i.e., constraint is binding) or at the EOQ value (i.e., the constraint is not binding)

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# Example

A retailer purchases an item from a supplier. Suppose that the item costs $30/unit to procure and the retailer's fixed order cost is $500/order. For this item, the retailer faces an annual demand of 24,000 units/year. Suppose that an annual inventory carrying rate of 0.25 is used by the retailer.

Due to a critical raw material shortage the supplier has recently announced that the maximum allowed order size 1,200 units.

a. What is the optimal order quantity for the retailer?
b. What is the retailer's average annual total (production, production setup and inventory carrying) cost associated with this item?
c. What is the financial impact of the supplier's maximum order size restriction on retailer's business?

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# Solution: Parameters

| | |
|---|---|
| Fixed ordering cost: | $K = \$500/\text{setup}$ |
| Unit production cost: | $c = \$30/\text{unit}$ |
| Ann. inventory carrying rate: | $i = 0.25/\text{year}$ |
| Inventory holding cost: | $h = ic = (0.25)(30) = \$7.5\ \text{unit/year}$ |
| Annual demand rate: | $\lambda = 24{,}000\ \text{units/year}$ |

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# Solution: Part (a)

$$q^* = \sqrt{\frac{2K\lambda}{h}} = \sqrt{\frac{2(500)(24{,}000)}{(0.25)(30)}} = \sqrt{\frac{2(500)(24{,}000)}{(7.5)}}$$

$$= 1{,}788.85 \text{ units}$$

We need to compare the retailer's EOQ against supplier's maximum order size restriction, to determine the retailer's optimal order quantity:

$$q^* = \min\{1200, 1788.85\} = 1{,}200 \text{ units}$$

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# Solution: Part (b)

$$C(q) = c\lambda + \frac{K\lambda}{q} + \frac{hq}{2}$$

$$= (30)(24{,}000) + \frac{(500)(24{,}000)}{1{,}200} + \frac{(7.5)(1{,}200)}{2}$$

$$= 720{,}000 + 10{,}000 + 4{,}500$$

$$= \$734{,}500/\text{year}$$

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# Solution: Part (c)

**Maximum size constraint**

$$q^* = \min\{C_U, q^*\} = 1{,}200 \text{ units}$$

$$C(q^*) = c\lambda + \frac{K\lambda}{q^*} + \frac{hq^*}{2}$$
$$= 720{,}000 + 10{,}000 + 4{,}500$$

$$C(q^*_{EPQ}) = \$734{,}500/\text{year}$$

**EOQ**

$$q^*_{EOQ} = \sqrt{\frac{2K\lambda}{h}} = 1{,}788.85 \text{ units}$$

$$C(q^*_{EOQ}) = c\lambda + \frac{K\lambda}{q^*_{EOQ}} + \frac{hq^*_{EOQ}}{2}$$
$$= 720{,}000 + 6{,}708.22 + 6708.19$$

$$C(q^*_{EOQ}) = \$733{,}416.41/\text{year}$$

Financial impact assessment including purchase cost

$$\frac{734{,}500 - 733{,}416.41}{733{,}416.71} = 0.0015 \equiv 0.15\%$$

Financial impact assessment excluding purchase cost

$$\frac{14{,}500 - 13{,}416.41}{13{,}416.71} = 0.0807 \equiv 8.07\%$$

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# EOQ: Modeling assumptions

1. The inventory level is reviewed continuously.
2. The planning horizon is very long (i.e., infinite), and the demand parameter will continue to be at the same value throughout this planning horizon.
3. The demand arrives at a known, constant, and continuous rate over time.
4. The entire order quantity can be delivered at the same time.
5. The order quantity need not be an integral number of units, and there are no minimum or maximum restrictions on the order quantity.
6. Although the item of interest can be an item in a larger inventory system, it is treated entirely independently of other items (i.e., a single-item system).
7. The cost parameters do not change over time. In particular, the inflation rate is low.
8. **The unit variable purchase cost is constant and does not depend on order quantity.**
9. The order delivery lead time is zero, i.e., instantaneous replenishment.
10. No shortages are allowed, i.e., all the demand must be satisfied.

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# Target.com

6 mega rolls $7.39
9 mega rolls $9.99
12 mega rolls $12.99
18 mega rolls $17.89

$1.23
$1.11
$1.08
$0.99

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# Discounting schemes

- We will restrict our attention to the most common types of discount structures
  - All-units discount scheme
  - Incremental discount scheme

- Discount pricing is also used in the context of transportation costs, which is known as the truckload discount scheme

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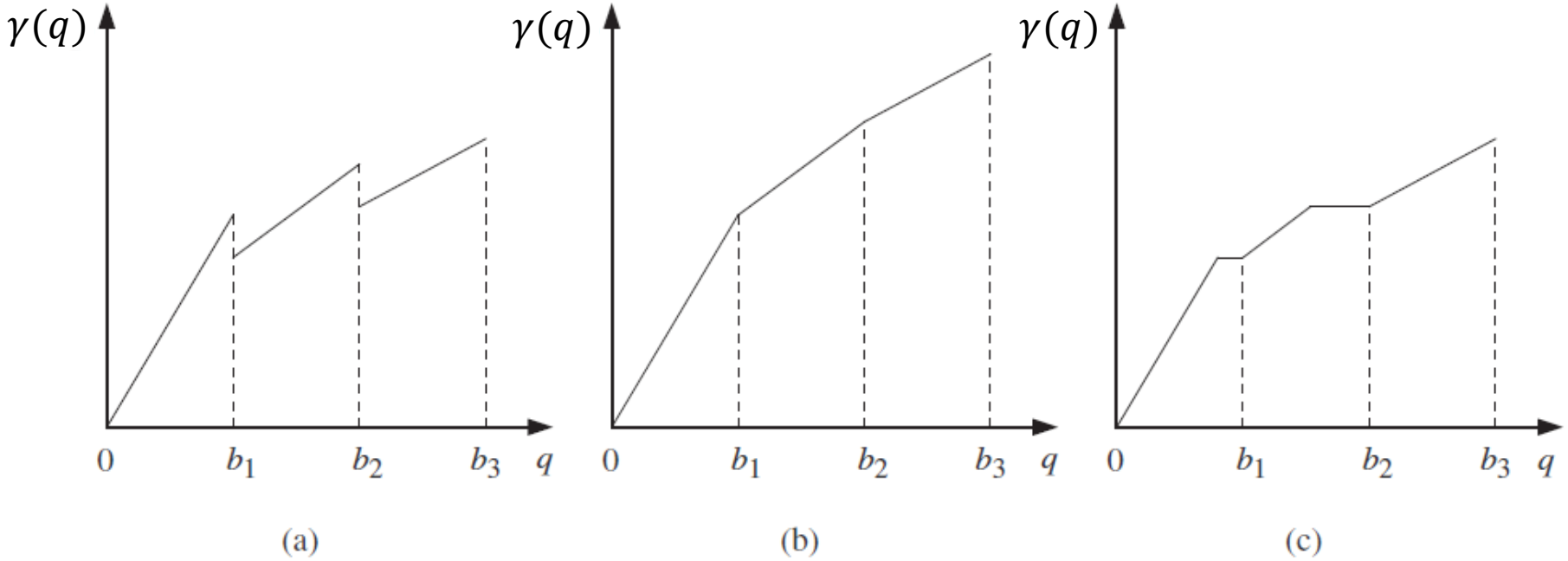
# Total purchase price




All units Incremental Truckload

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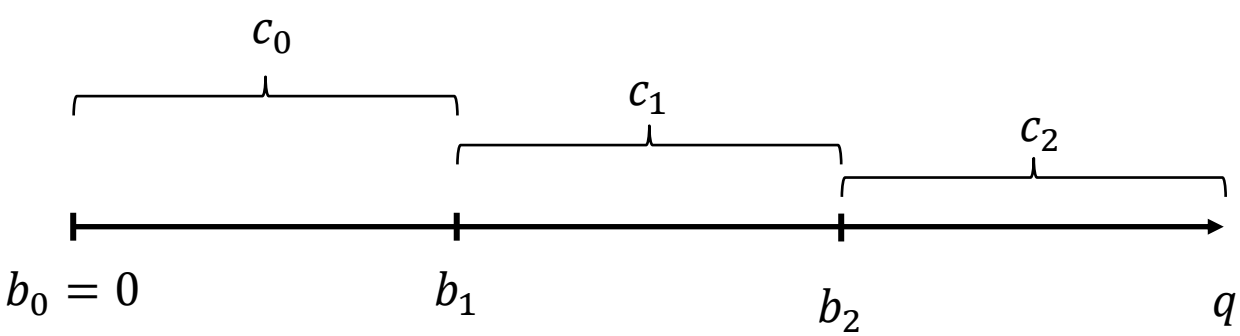
# All-units discount scheme

All-units discount scheme is characterized by a set of break points, indexed by $j = 0, \dots, r$, $b_0, b_1, \dots, b_r$, where

1. $b_0 < b_1 < \cdots < b_r$ and
2. $b_0 = 0$

and a set of unit prices $c_0, c_1, \dots, c_r$ where

$$c_0 > c_1 > \cdots > c_r$$

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# All-units discount scheme

- All-units discount scheme can be defined by a set of break-points and a set of unit prices.

- If the order size exceeds the break-point quantity then the discounted unit price is applied to all the units in the order.

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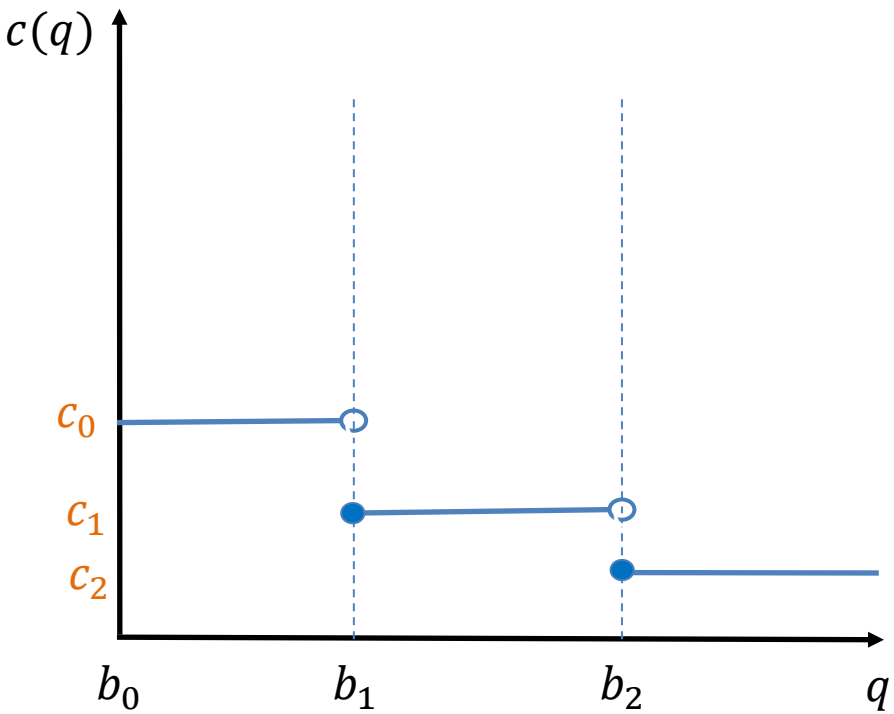
# All-units: Unit procurement cost




Unit procurement cost, $c(q)$

$$c(q) = \begin{cases} c_0 & 0 \leq q < b_1 \\ c_1 & b_1 \leq q < b_2 \\ c_2 & b_2 \leq q < \infty \end{cases}$$

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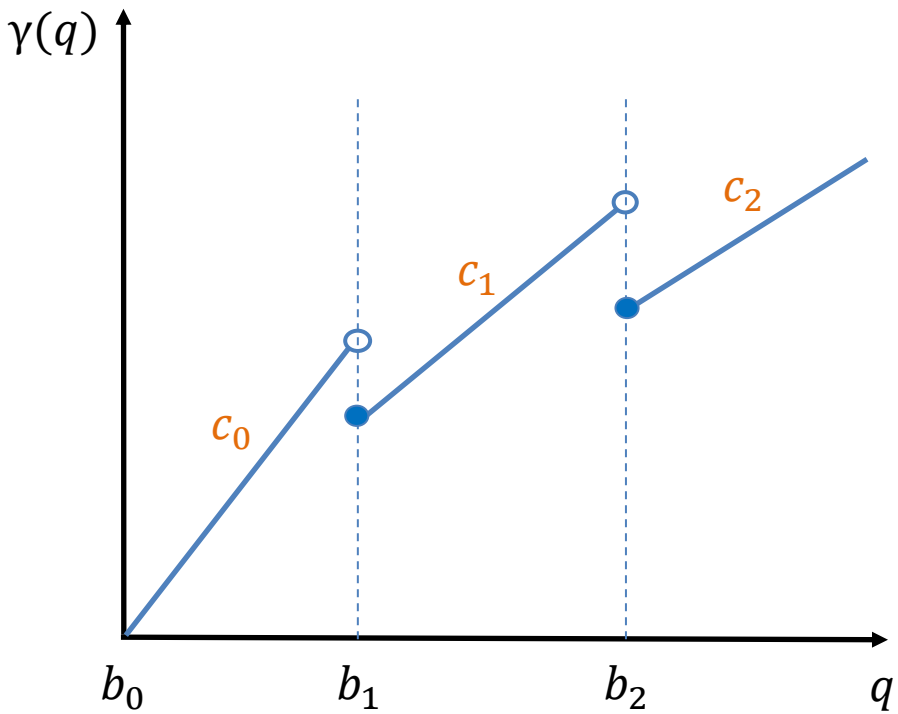
# All-units: Total procurement cost




Total procurement cost, $\gamma(q)$

$$\gamma(q) = \begin{cases} c_0 q & 0 \le q < b_1 \\ c_1 q & b_1 \le q < b_2 \\ c_2 q & b_2 \le q < \infty \end{cases}$$

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# All-units: Average annual total cost

The average annual total cost of procurement, ordering and inventory holding is a function of the order quantity $q$ and the corresponding unit procurement price $c(q)$ at a particular price level:

$$C(q, c(q)) = \begin{cases} c_0\lambda + \dfrac{K\lambda}{q} + \dfrac{ic_0 q}{2} & 0 \leq q < b_1 \\ c_1\lambda + \dfrac{K\lambda}{q} + \dfrac{ic_1 q}{2} & b_1 \leq q < b_2 \\ c_2\lambda + \dfrac{K\lambda}{q} + \dfrac{ic_2 q}{2} & b_2 \leq q < \infty \end{cases}$$

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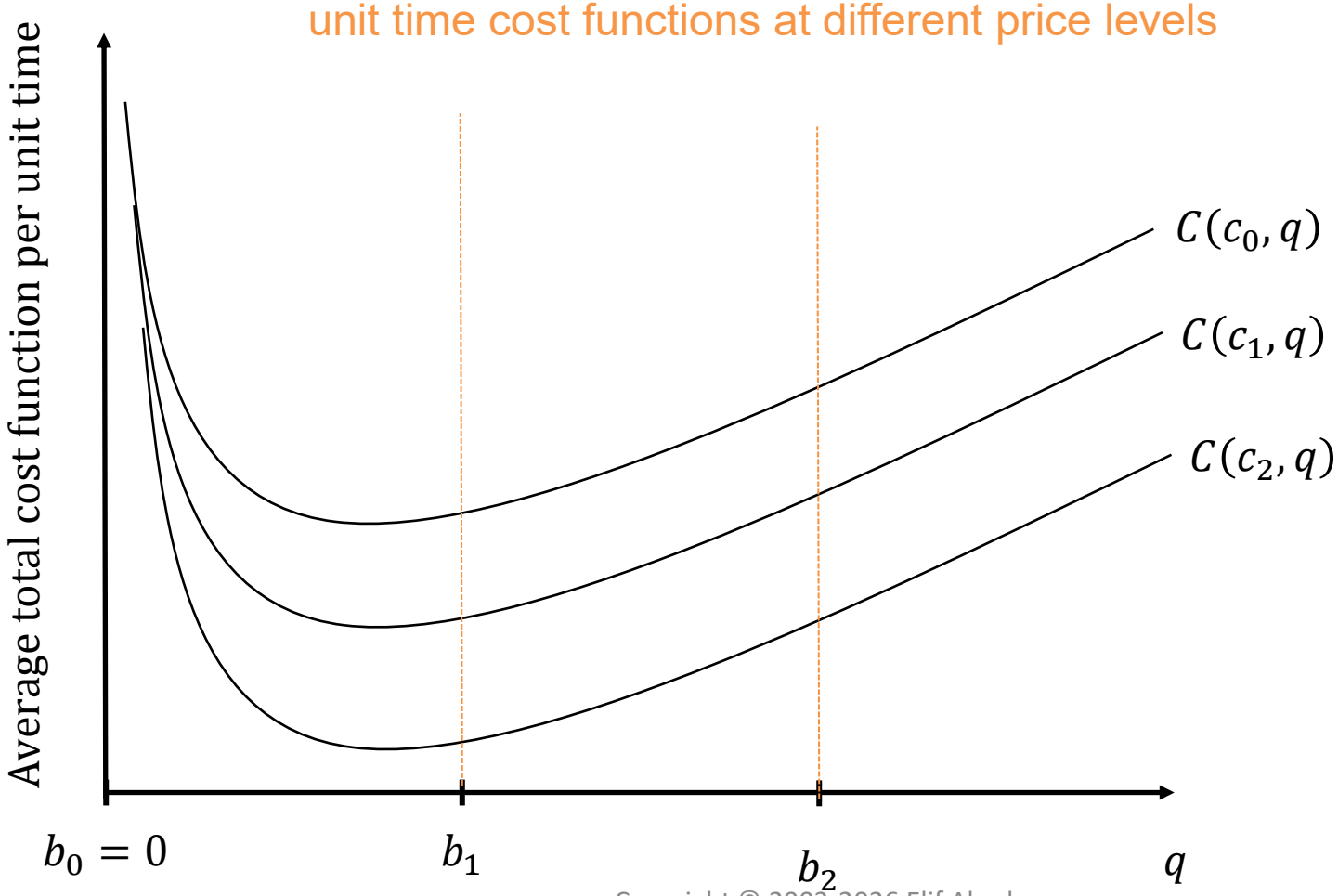
# All-units: Total cost function

Relative positions of average annual total cost per unit time cost functions at different price levels

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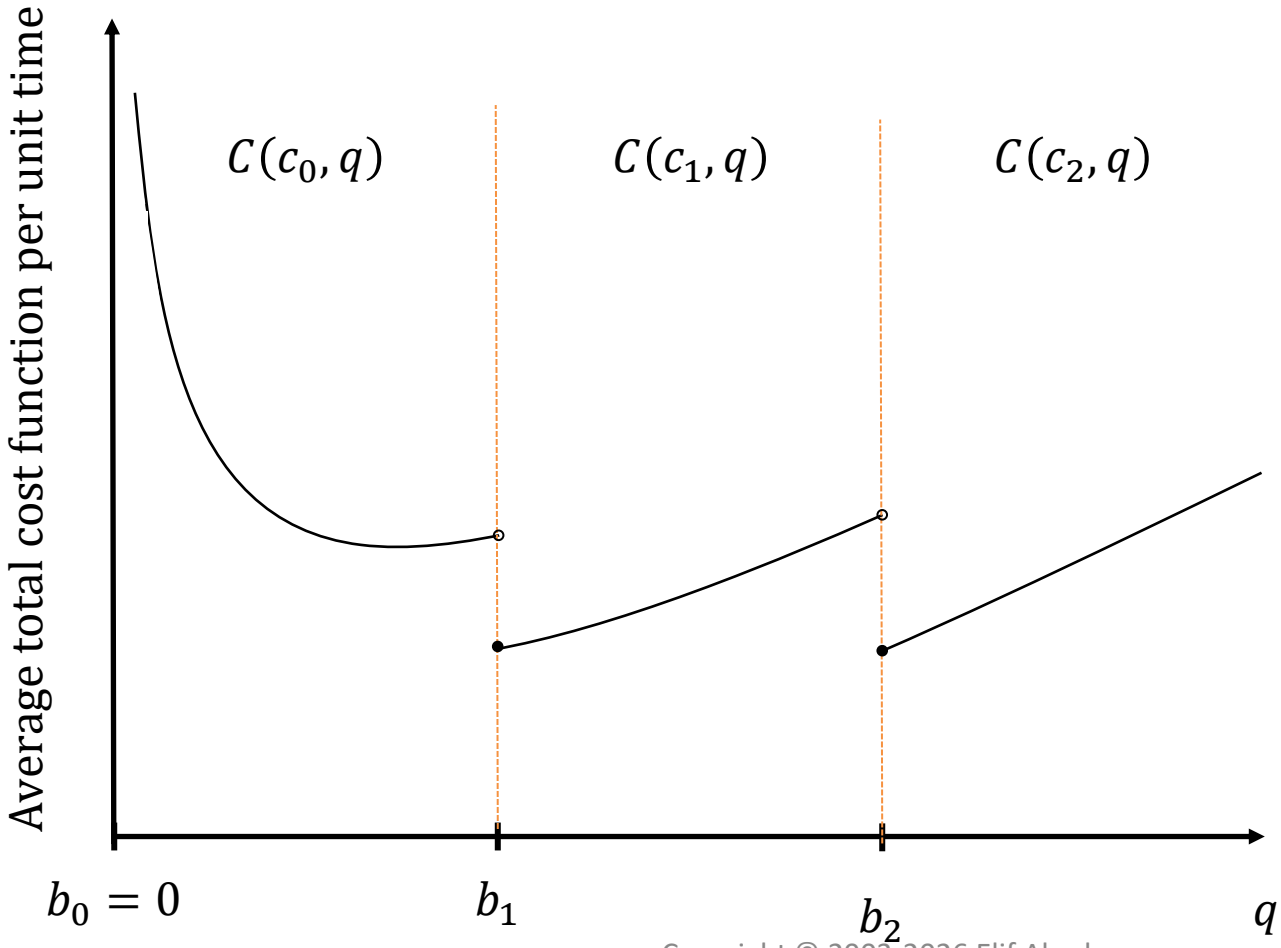
# All-units: Total cost function

Overall feasible average annual total cost per unit time cost function

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# All-units: Approach: Overall idea

Observation: The average annual total cost function is piece-wise convex for each price level and discontinuous at breakpoints

Approach: An intuitive sketch of the approach is as follows:

1. For each price level, we can plot the EOQ cost curve and calculate the EOQ value. The question is whether this EOQ value is feasible. If it is not feasible, we need to think about the local optimal solution for each price level.
2. Identify the local optimal (i.e., *candidate*) solution for each price level, if possible.
3. Choose from among all local optimal (i.e., candidate) solutions to determine the global optimal solution.

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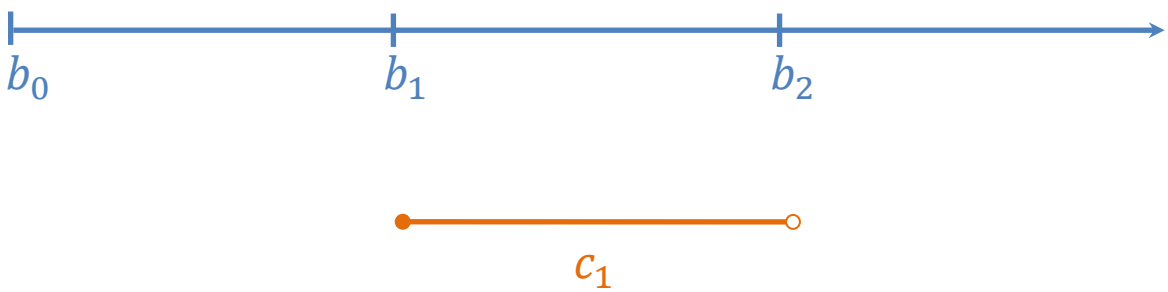
# All-units: Candidate solutions

Given the entire feasible region, let's consider a portion:




Where can the local optimal (i.e., candidate) solution be for a given range?

It can never be at the upper bound!!

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# All-units: Approach (1 of 2)

**Step 1:** Evaluate the EOQ value for each price level $c_j$, $0 \leq j \leq r$, using

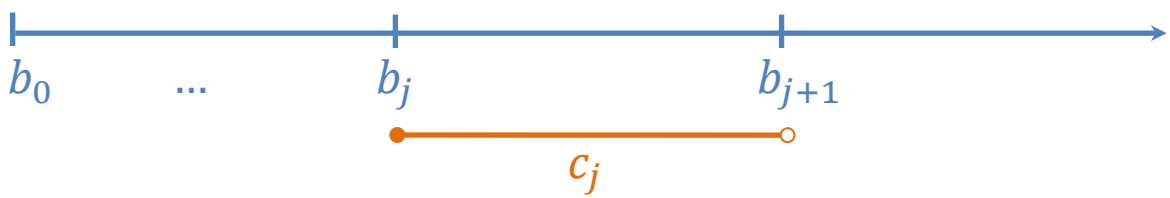
$$q_j^{EOQ} = \sqrt{\frac{2K\lambda}{ic_j}} \text{ for } j = 1, \dots, n$$

**Step 2:** Using each EOQ values from Step 1, identify the candidate solution (if there is one) for each range, using the corresponding lower $b_j$ and upper bound $b_{j+1}$ for the range:

<u>Case 1</u>: If $q_j^{EOQ} < b_j$, then $q_j^* = b_j$.

<u>Case 2</u>: If $b_j \leq q_j^{EOQ} < b_{j+1}$, then $q_j^* = q_j^{EOQ}$.

<u>Case 3</u>: If $b_{j+1} < q_j^{EOQ}$, there is no candidate (i.e., local optimal) solution.

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# All-units: Approach (2 of 2)

**Step 3:** For each $j$, with a candidate (i.e., local optimal) solution, evaluate the average annual total cost using

$$C(q_j^*) = \lambda c_j + \frac{\lambda K}{q_j^*} + \frac{i c_j q_j^*}{2} \text{ for } j = 1, \dots, n$$

**Step 4:** Over all values of $j$, select the order quantity $q_j^*$ that yields the lowest average annual total cost.

**For this problem variant, you MUST include the total procurement cost in the total cost calculation, as that's where most of the cost savings may reside.**

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# Example: All-units (1 of 4)

Consider an item for which the following all-units discount scheme is in effect where $b_0 = 0, b_1 = 5{,}000$ and $b_2 = 10{,}000$ :

| Order Quantity | Unit Price |
|---|---|
| $0 \leq q < 5000$ | \$3.00 |
| $5{,}000 \leq q < 10{,}000$ | \$2.96 |
| $10{,}000 \leq q < \infty$ | \$2.92 |

For this item, the relatively stable demand is 10,000 units/month, the fixed ordering cost is \$100/order and an annual inventory carrying rate of 20% is being used.

What is the optimal order quantity for this item?

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# Example: All-units (2 of 4)

**Step 1:** Evaluate the EOQ for each price level $c_j$, $0 \leq j \leq r$

$$q_0^{EOQ} = \sqrt{\frac{(2)(100)(120{,}000)}{(0.20)(3)}} = 6{,}324.56 \text{ units}$$

$$q_1^{EOQ} = \sqrt{\frac{(2)(100)(120{,}000)}{(0.20)(2.96)}} = 6{,}367.15 \text{ units}$$

$$q_2^{EOQ} = \sqrt{\frac{(2)(100)(120{,}000)}{(0.20)(2.92)}} = 6{,}410.61 \text{ units}$$

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# Example: All-units (3 of 4)

<u>*Case 1*</u>: If $q_j^{EOQ} < b_j$, then $q_j^* = b_j$.
<u>*Case 2*</u>: If $b_j \leq q_j^{EOQ} < b_{j+1}$, then $q_j^* = q_j^{EOQ}$.
<u>*Case 3*</u>: If $b_{j+1} < q_j^{EOQ}$, there is no candidate solution.

$q_0^{EOQ} = 6{,}324.56$ units; Compare against the range $0 \leq q < 5000$;
Case 3 above applies (larger than the upper bound);
Do nothing.

$q_1^{EOQ} = 6{,}367.15$ units; Compare against the range $5{,}000 \leq q < 10{,}000$;
Case 2 above applies (between the lower and upper bounds);
$q_1^* = 6{,}367.15$ units.

$q_2^{EOQ} = 6{,}411.61$ units; Compare against the range $10{,}000 \leq q < \infty$;
Case 1 above applies (less than the lower bound);
$q_2^* = 10{,}000$ units.

Note that, for this example problem, each one of the Cases 1, 2 and 3 applies only once.

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# Example: All-units (1 of 4)

**Step 3:** Evaluate average annual total cost

$$C(6{,}367.15, 2.96) = (120{,}000)(2.96) + \frac{(100)(120{,}000)}{6{,}367.15} + \frac{(0.20)(2.96)(6{,}367.15)}{2}$$
$$= \$358{,}969$$

$$C(10{,}000, 2.92) = (120{,}000)(2.92) + \frac{(100)(120{,}000)}{10{,}000} + \frac{(0.20)(2.92)(10{,}000)}{2}$$
$$= \boxed{\$354{,}520}$$

**Step 4:** Identify the optimal decision: Order 10,000 units/order at $2.92/unit.

Note: In this problem, it is a pure coincidence that the monthly demand is equal to one of the breakpoints which also turns out to be the optimal order quantity.

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# Incremental discount scheme

- Incremental discount scheme can also be defined by a set of break-points along with a base price and a discount rate or a set of discount prices.

- If the order size exceeds the break-point quantity then the discount rate is applied only to those units that are beyond the breakpoint.

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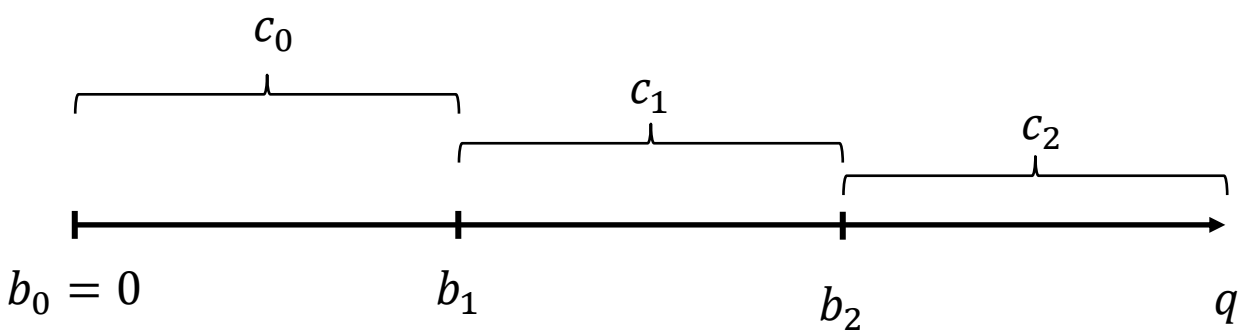
# Incremental discount scheme

Similar to an all-unit discount scheme, an incremental discount scheme is also characterized by a set of break points, indexed by $j = 0, \dots, r$, $b_0, b_1, \dots, b_r$, where

1. $b_0 < b_1 < \cdots < b_r$ and
2. $b_0 = 0$

along with a set of unit prices $c_0, c_1, \dots, c_r$ where

$$c_0 > c_1 > \cdots > c_r$$

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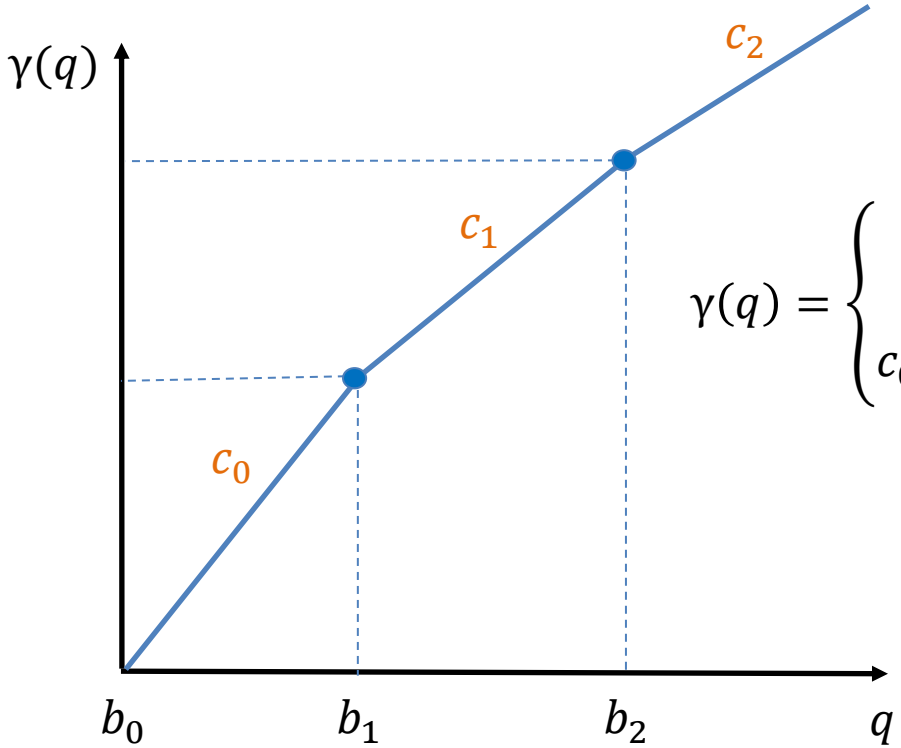
# Incremental: Total procurement cost




Total procurement cost, $\gamma(q)$

$$\gamma(q) = \begin{cases} c_0 q & 0 \le q < b_1 \\ c_0 b_1 + c_1(q - b_1) & b_1 \le q < b_2 \\ c_0 b_1 + c_1(b_1 - b_1) + c_2(q - b_2) & b_2 \le q < \infty \end{cases}$$

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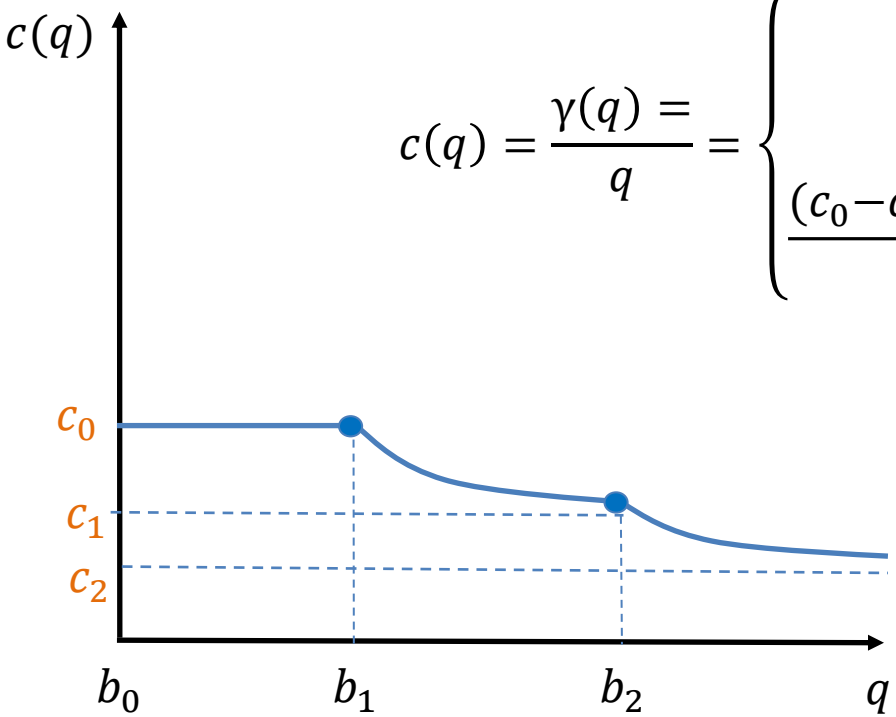
# Incremental: Unit procurement cost

Unit procurement cost, $c(q)$

$$c(q) = \frac{\gamma(q) =}{q} = \begin{cases} c_0 & 0 \le q < b_1 \\ \dfrac{(c_0 - c_1)b_1}{q} + c_1 & b_1 \le q < b_2 \\ \dfrac{(c_0 - c_1)b_1 + (c_1 - c_2)b_1}{q} + c_2 & b_2 \le q < \infty \end{cases}$$

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# Insight: Incremental

The average annual total cost of procurement, ordering and inventory holding is a function of the order quantity $q$ and the corresponding effective unit procurement price $c'(q)$ in a particular price range:

$$C(q, c(q)) = \begin{cases} c_0\lambda + \dfrac{K\lambda}{q} + \dfrac{ic_0 q}{2} & 0 \le q < b_1 \\ \left(\dfrac{(c_0 - c_1)b_1}{q} + c_1\right)\lambda + \dfrac{K\lambda}{q} + \dfrac{i\left(\dfrac{(c_0 - c_1)b_1}{q} + c_1\right)q}{2} & b_1 \le q < b_2 \\ \left(\dfrac{(c_0 - c_1)b_1 + (c_1 - c_2)b_1}{q} + c_2\right)\lambda + \dfrac{K\lambda}{q} + \dfrac{i\left(\dfrac{(c_0 - c_1)b_1 + (c_1 - c_2)b_1}{q} + c_2\right)q}{2} & b_2 \le q < \infty \end{cases}$$

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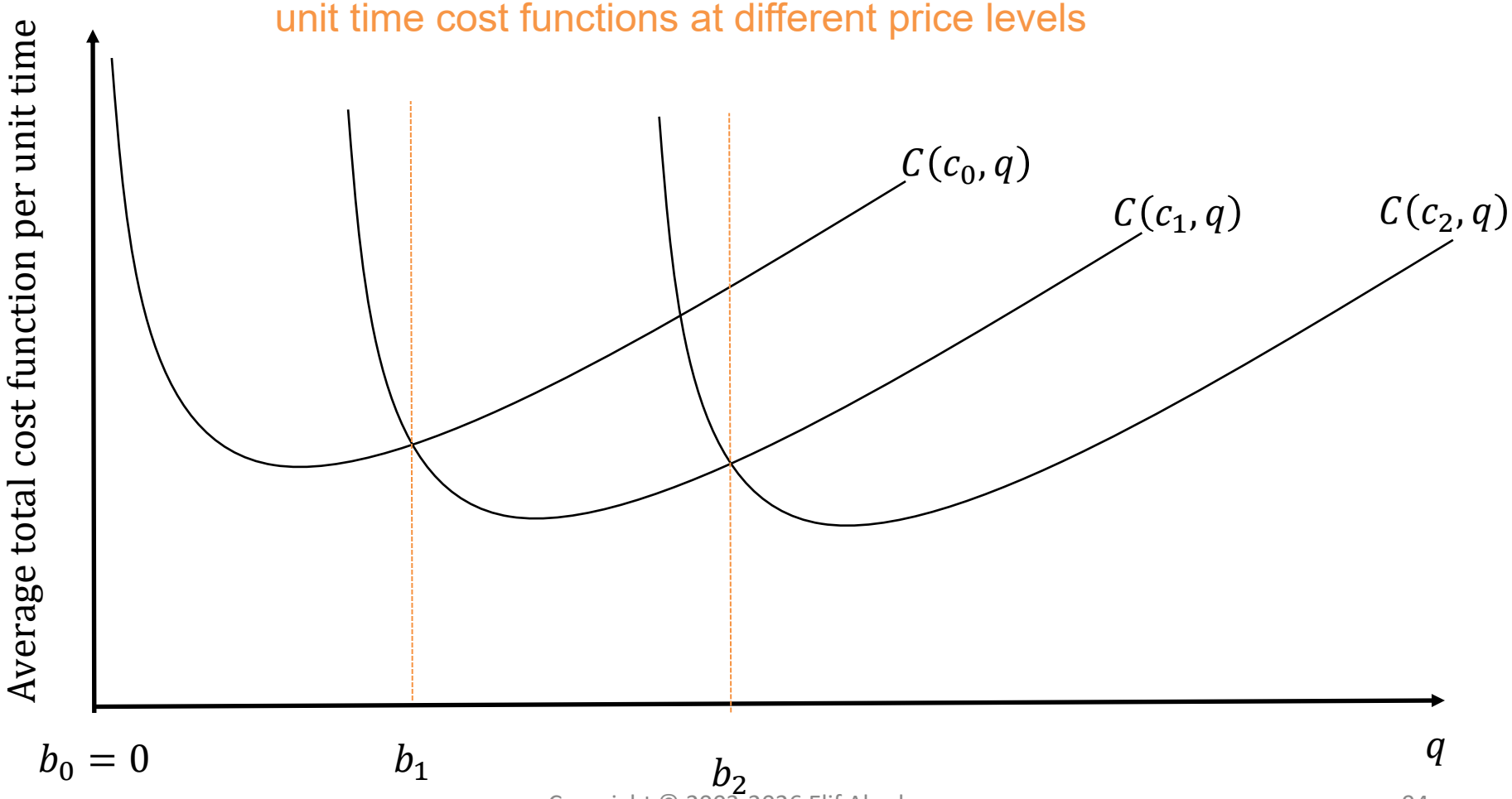
# Incremental: Total cost function

Relative positions of average annual total cost per unit time cost functions at different price levels

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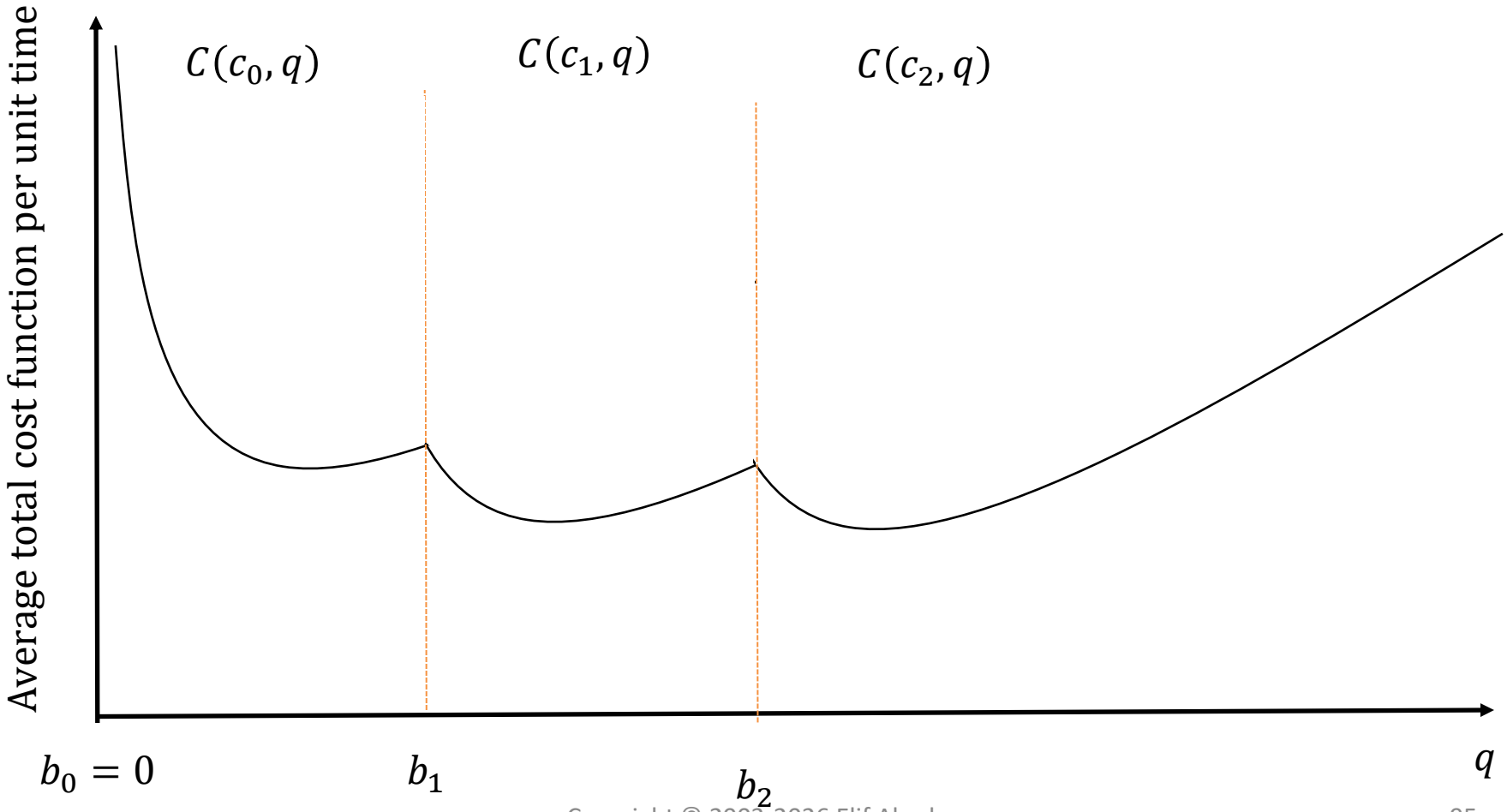
# Incremental: Total cost function

Overall feasible average annual total cost per unit time cost function

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# Incremental: Approach: Overall idea

- The overall cost function is piece-wise convex but discontinuous at break points
- Identify a candidate solution for each range if possible
- Choose from among all candidate solutions.

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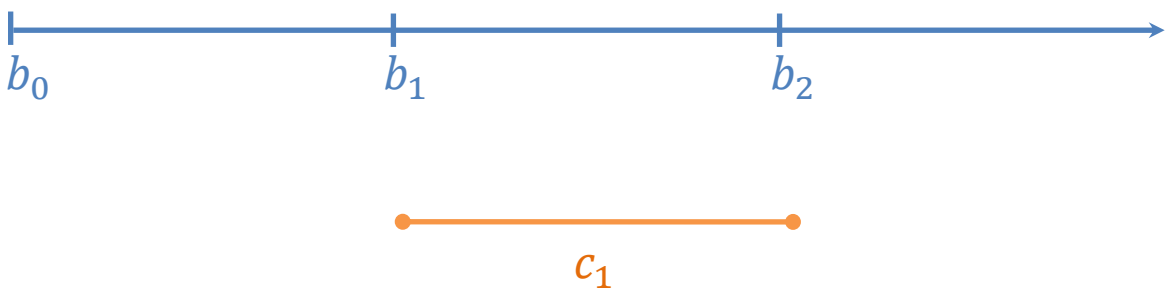
# Incremental: Candidate solutions

Given the entire feasible region, let's consider a portion:




Where can the candidate solution be for a given range?

It can be anywhere in the range including the upper bound, since two consecutive cost functions intersect at the upper bound!

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# Approach: Incremental (1 of 3)

Let $V_j$ denote the cost of ordering $b_j$ units.

Define $V_0 = 0$ and $V_j$ for $0 \leq j \leq r$ is given by

$$V_j = c_0(b_1 - b_0) + c_1(b_2 - b_1) + \cdots + c_{j-1}\left(b_j - b_{j-1}\right)$$

where $r$ is the last breakpoint.

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# Approach: Incremental (2 of 3)

**Step 1:** Evaluate EOQ values for each $j$ using

$$q_j^{EOQ} = \sqrt{\frac{2\lambda\left(K + V_j - b_j c_j\right)}{i c_j}} \text{ for } j = 1, \dots, r.$$

**Step 2:** Using EOQ values from Step 1, identify the candidate solution for each range, using the corresponding lower $b_j$ and upper bound $b_{j+1}$ for the range:

<u>*Case 1*</u>: If $q_j^{EOQ} < b_j$ then $q_j^* = b_j$ .

<u>*Case 2*</u>: If $b_j \leq q_j^{EOQ} < b_{j+1}$ then $q_j^* = q_j^{EOQ}$.

<u>*Case 3*</u>: If $b_{j+1} < q_j^{EOQ}$ then $q_j^* = b_{j+1}$ .

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# Approach: Incremental (3 of 3)

**Step 3:** For each $j$, calculate the total annual cost of ordering using

$$C_j(q_j^*) = \frac{K\lambda}{q_j^*} + \frac{i}{2}\left[V_j + (q_j^* - b_j)c_j\right] + \frac{\lambda}{q_j^*}\left[V_j + (q_j^* - b_j)c_j\right]$$

$$= \frac{K\lambda}{q_j^*} + \left(\frac{i}{2} + \frac{\lambda}{q_j^*}\right)\left[V_j + (q_j^* - b_j)c_j\right]$$

**Step 4:** Over all values of $j$, select the order size $q_j^*$ that yields the lowest average annual total cost.

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# Example: Incremental

Consider an item for which the following incremental discounting scheme is in effect $b_0 = 0, b_1 = 5{,}000$ and $b_2 = 10{,}000$

| Order Quantity | Unit Price |
|---|---|
| 0 < $q$ < 5,000 | \$3.00 |
| 5,000 ≤ $q$ < 10,000 | \$2.96 |
| 10,000 ≤ $q$ | \$2.92 |

For this item, the demand is 10,000 units/month, the fixed ordering cost is \$100/order and an annual inventory carrying rate of 20% is being used.

What is the optimal ordering quantity for this item?

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# Example: Incremental

**Initialization:** We have

$$V_0 = \$0$$
$$V_1 = (3)(5{,}000 - 0) = \$15{,}000$$
$$V_2 = (3)(5{,}000 - 0) + (2.96)(10{,}000 - 5{,}000) = \$29{,}800$$

**Step 1:** We have

$$q_0^{EOQ} = \sqrt{\frac{2\lambda(K+V_0-b_0c_0)}{ic_0}} = \sqrt{\frac{2(120{,}000)(100+0-(0)(3))}{(0.2)(3)}} = 6{,}324.55 \text{ units}$$

$$q_1^{EOQ} = \sqrt{\frac{2\lambda(K+V_1-b_1c_1)}{ic_1}} = \sqrt{\frac{2(120{,}000)(100+15{,}000-(5{,}000)(2.96))}{(0.2)(2.96)}} = 11{,}028.22 \text{ units}$$

$$q_2^{EOQ} = \sqrt{\frac{2\lambda(K+V_2-b_2c_2)}{ic_2}} = \sqrt{\frac{2(120{,}000)(100+29{,}800-(10{,}000)(2.92))}{(0.2)(2.92)}} = 16{,}960.87 \text{ units}$$

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# Example: Incremental

Case 1: If $q_j^{EOQ} < b_j$ then $q_j^* = b_j$ .

Case 2: If $b_j \leq q_j^{EOQ} < b_{j+1}$ then $q_j^* = q_j^{EOQ}$.

Case 3: If $b_{j+1} < q_j^{EOQ}$ then $q_j^* = b_{j+1}$ .

**Step 3:** We have

$q_0^{EOQ} = 6{,}324.56$ units; Compare against the range $0 \leq q < 5000$;
Case 3 above applies (larger than the upper bound);
$q_0^* = b_1 = 5{,}000$.

$q_1^{EOQ} = 11{,}028.22$units; Compare against the range $5{,}000 \leq q < 10{,}000$;
Case 3 above applies (larger than the upper bound);
$q_1^* = b_2 = 10{,}000$ units.

$q_2^{EOQ} = 16{,}960.87$units; Compare against the range $10{,}000 \leq q < \infty$;
Case 2 above applies (between the bounds);
$q_2^* = 16{,}960.87$units.

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# Example: Incremental

$$C_j(q_j^*) = \frac{K\lambda}{q_j^*} + \left(\frac{i}{2} + \frac{\lambda}{q_j^*}\right)[V_j + (q_j^* - b_j)c_j]$$

**Step 3:** We have

$$C(5{,}000) = \frac{(100)(120{,}000)}{5{,}000} + \left(\frac{0.2}{2} + \frac{120{,}000}{5{,}000}\right)[0 + (5{,}000 - 0)(3.00)] = \$363{,}900$$

$$C(10{,}000) = \frac{(100)(120{,}000)}{10{,}000} + \left(\frac{0.2}{2} + \frac{120{,}000}{10{,}000}\right)[15{,}000 + (10{,}000 - 5{,}000)(2.96)] = \$361{,}780$$

$$C(16{,}960.87) = \frac{(100)(120{,}000)}{16{,}960.87} + \left(\frac{0.2}{2} + \frac{120{,}000}{16{,}960.87}\right)[29{,}800 + (16{,}960.87 - 10{,}000)(2.92)] = \boxed{\$360{,}365}$$

**Step 4:** Identify the optimal decision: Order 16,960.87 units/order such that 5,000 units at $3.00/unit, another 5,000 units at $2.96/unit, 6,960.87 at $2.92/unit.

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# Example: Incremental

## All-units discounting

Order 10,000 units/order at $2.92/unit.

$$C(10{,}000) = \$354{,}520$$

## Incremental discounting

Order 16,960.87 units/order such that 5,000 units at $3.00/unit, another 5,000 units at $2.96/unit, 6,960.87 at $2.92/unit.

$$C(16{,}960.87) = \$360{,}365$$

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# EOQ: Modeling assumptions

1. The inventory level is reviewed continuously.
2. The planning horizon is very long (i.e., infinite), and the demand parameter will continue to be at the same value throughout this planning horizon.
3. The demand arrives at a known, constant, and continuous rate over time.
4. The entire order quantity can be delivered at the same time.
5. The order quantity need not be an integral number of units, and there are no minimum or maximum restrictions on the order quantity.
6. **Although the item of interest can be an item in a larger inventory system, it is treated entirely independently of other items (i.e., a single-item system).**
7. The cost parameters do not change over time. In particular, the inflation rate is low.
8. The unit variable purchase cost is constant and does not depend on order quantity.
9. The order delivery lead time is zero, i.e., instantaneous replenishment.
10. No shortages are allowed, i.e., all the demand must be satisfied.

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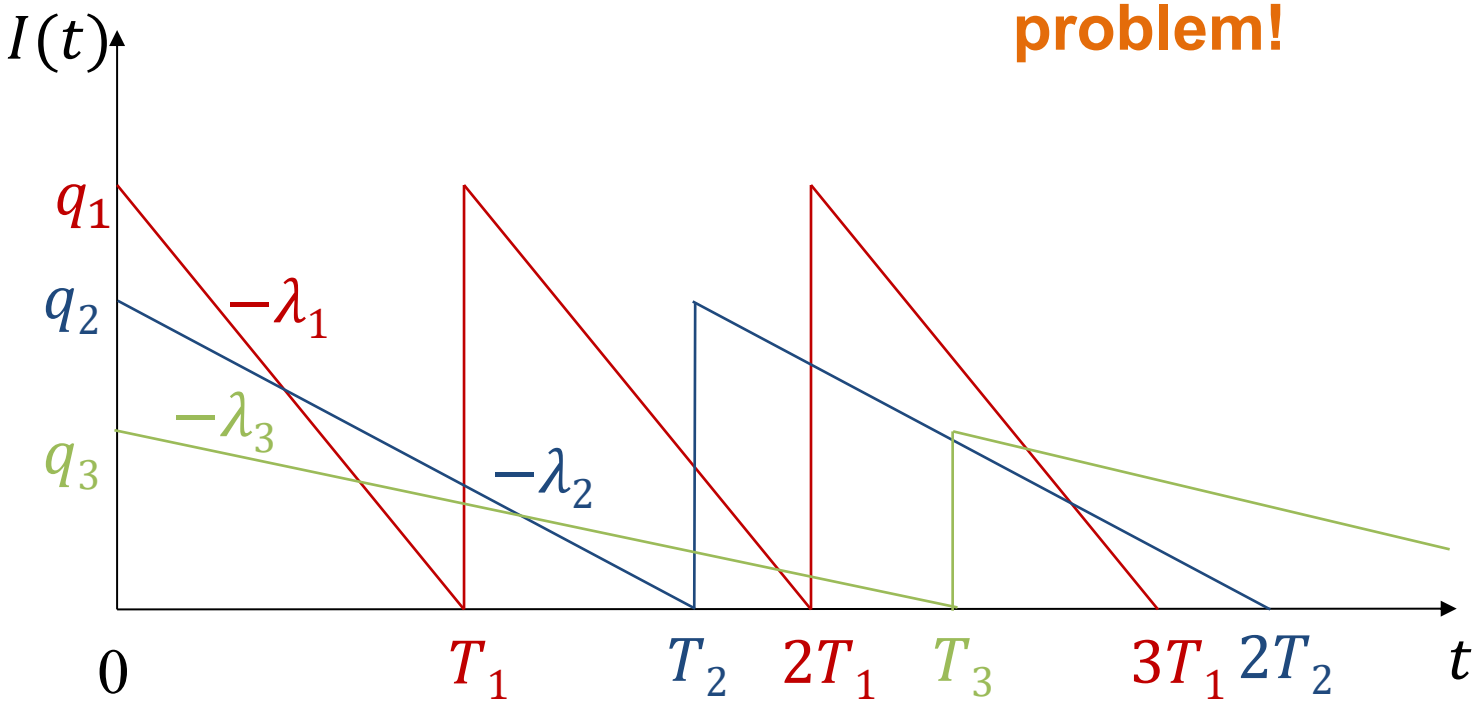
# Lot sizing for multiple products

**Complicated control problem!**

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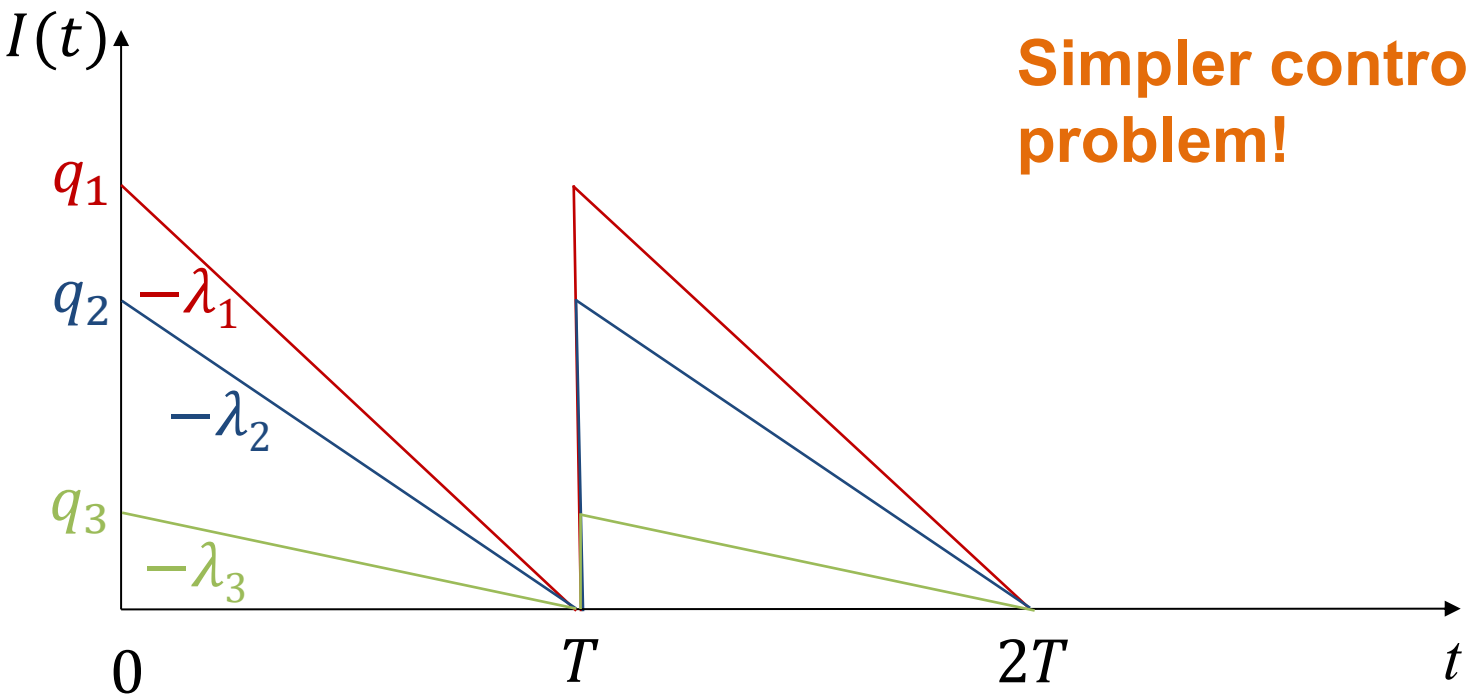
# Multiple products




**Simpler control problem!**

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# Multiple products: Individual ordering

- A set of items indexed by $j = 1, \dots, n$ with demand $\lambda_j$, unit purchase cost $c_j$
- Each item incurs a fixed ordering cost of $K_j$ for $j = 1, \dots, n$
- All items have their own cycle length $T_j$ for $j = 1, \dots, n$

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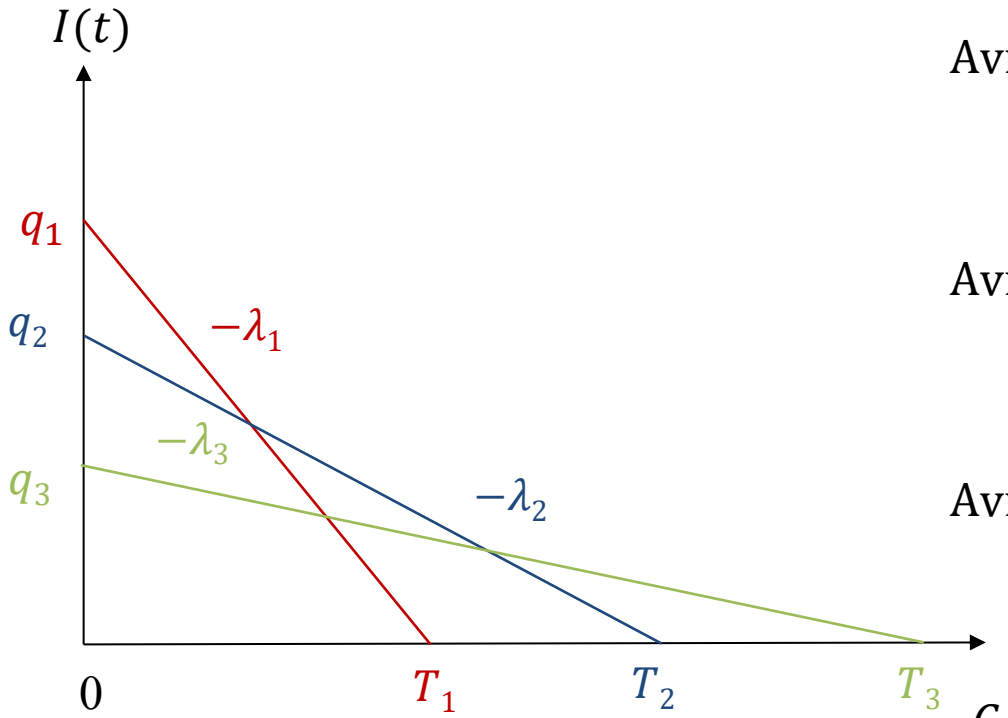
# Multiple products: Individual ordering




$$\text{Avr. Ann. Procurement Cost} = \sum_{j=1}^{n} c_j \lambda_j$$

$$\text{Avr. Ann. Ordering Cost} = \sum_{j=1}^{n} \frac{K_j \lambda_j}{q_j}$$

$$\text{Avr. Ann. Inventory Holding Cost} = \sum_{j=1}^{n} \frac{i c_j q_j}{2}$$

$$C_I(q_1, q_2, \dots, q_n) = \sum_{j=1}^{n} \left( c_j \lambda_j + \frac{K_j \lambda_j}{q_j} + \frac{i c_j q_j}{2} \right)$$

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# Multiple products: Individual ordering: Optimal solution

$C_I(q_1, q_2, \dots, q_n)$ is separable for each item. Thus, the optimal order quantity for item $j$ is given by

$$q_j^* = \sqrt{\frac{2K_j\lambda_j}{ic_j}}$$

Hence, the total cost is given by

$$C_I(q_1^*, q_2^*, \dots, q_n^*) = \sum_{j=1}^{n}\left(c_j\lambda_j + \frac{K_j\lambda_j}{q_j^*} + \frac{ic_jq_j^*}{2}\right)$$

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# Multiple products: Joint ordering

- A set of items indexed by $j = 1, \ldots, n$ with demand $\lambda_j$ and unit purchase cost $c_j$

- A fixed shared cost $K$ incurred each time an order is placed independent of the variety of products included in the order

- All items share the same cycle length $T$

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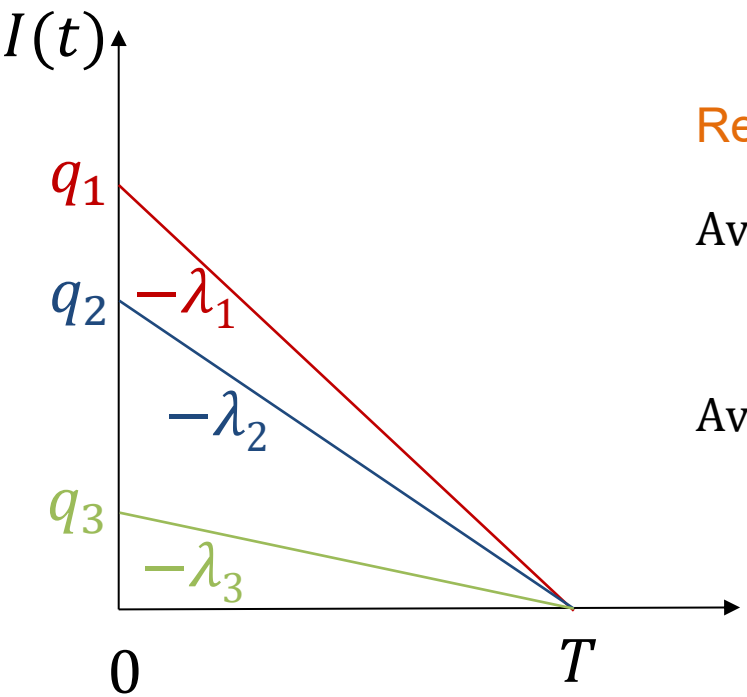
# Multiple products: Joint ordering




$$\text{Avr. Ann. Procurement Cost} = \sum_{j=1}^{n} c_j \lambda_j$$

Recall that $q_j = \lambda_j T$

$$\text{Avr. Ann. Ordering Cost} = \frac{K}{T}$$

$$\text{Avr. Ann. Inventory Holding Cost} = \sum_{j=1}^{n} \frac{i c_j \lambda_j T}{2}$$

$$C_J(T) = \frac{K}{T} + \sum_{j=1}^{n} \left( c_j \lambda_j + \frac{i c_j \lambda_j T}{2} \right)$$

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# Multiple products: Joint ordering: Optimal solution

$C_J(T)$ has a similar functional form to that of the basic EOQ model. Thus, the optimal cycle length given by

$$T^* = \sqrt{\frac{2K}{\sum_{j=1}^{n} i c_j \lambda_j}}$$

Hence, the total cost is given by

$$C_J(T^*) = \frac{K}{T^*} + \sum_{j=1}^{n} \left( c_j \lambda_j + \frac{i c_j \lambda_j T^*}{2} \right)$$

and the corresponding optimal order quantities are given by

$$q_j^* = \lambda_j T^*$$

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# Example

Consider two electronic components A and B. Annual demand rate for these items are 2,400 units and 1,200 units, respectively. Unit procurement costs are $2/unit and $5/unit, respectively also. The fixed cost of placing an order for item A is $30 whereas it is $40 for item B. Annual inventory carrying rate is 8 percent.

The supplier makes an offer to replenish these items jointly at a fixed ordering cost of $60 per order. Should the manufacturer accept this offer?

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# Solution: Individual ordering

$\lambda_A = 2{,}400$
$K_A = \$30/\text{order}$
$c_A = \$2/\text{unit}$
$i = 0.08$

Optimal order quantity for Component A:

$$q_A^* = \sqrt{\frac{2K_A\lambda_A}{ic_A}} = \sqrt{\frac{2(30)(2{,}400)}{(0.08)(2)}} = 948.68 \text{ units}$$

Optimal order quantity for Component B:

$\lambda_B = 1{,}200$
$K_B = \$40/\text{order}$
$c_B = \$5/\text{unit}$
$i = 0.08$

$$q_B^* = \sqrt{\frac{2K_B\lambda_B}{ic_B}} = \sqrt{\frac{2(40)(1{,}200)}{(0.08)(5)}} = 489.90 \text{ units}$$

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# Solution: Individual ordering

Average annual total cost for A and B under individual ordering:

$$C_I(q_A^*, q_B^*) = \sum_{j=1}^{j\in\{A,B\}} \left( c_j\lambda_j + \frac{K_j\lambda_j}{q_j^*} + \frac{ic_jq_j^*}{2} \right)$$

Average annual total cost of ordering and inventory holding for A and B under individual ordering:

$$C_I(q_A^*, q_B^*) = \sum_{j=1}^{j\in\{A,B\}} \left( \frac{K_j\lambda_j}{q_j^*} + \frac{ic_jq_j^*}{2} \right) = \frac{(30)(2400)}{948.60} + \frac{(0.08)(2)(948.60)}{2}$$
$$+ \frac{(40)(1200)}{489.90} + \frac{(0.08)(5)(489.90)}{2} = \$347.75/\text{year}$$

Average annual total cost including purchase cost:

$$\$347.75 + (2)(2{,}400) + (5)(1{,}200) = \$11{,}147.75/\text{year}$$

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# Solution: Joint ordering

$\lambda_A = 2{,}400$
$\lambda_B = 1{,}200$
$K = \$60/\text{order}$
$c_A = \$2/\text{unit}$
$c_B = \$5/\text{unit}$
$i = 0.08$

Shared cycle length (report 4 digits):

$$T^* = \sqrt{\frac{2K}{\sum_{j=1}^{j\epsilon\{A,B\}} i c_j \lambda_j}} = \sqrt{\frac{2(60)}{(0.08)(2)(2400) + (0.08)(5)(1200)}} = 0.3727\ years$$

Optimal order quantity for Component A:

$$q_A^* = \lambda_A T^* = (2{,}400)(0.3727) = 894.48\ units$$

Optimal order quantity for Component B:

$$q_B^* = \lambda_B T^* = (1{,}200)(0.3727) = 447.24\ units$$

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# Solution: joint ordering

Average annual total cost for A and B under joint ordering:

$$C_J(T^*) = \frac{K}{T^*} + \sum_{j=1}^{j\in\{A,B\}} \left( c_j\lambda_j + \frac{ic_j\lambda_j T^*}{2} \right)$$

Average annual total cost of ordering and inventory holding for A and B under joint ordering:

$$C_J(T^*) = \frac{K}{T^*} + \sum_{j=1}^{j\in\{A,B\}} \left( \frac{ic_j\lambda_j T^*}{2} \right) = \frac{60}{0.3727} + \frac{(0.08)(2)(2{,}400)(0.3727)}{2} + \frac{(0.08)(5)(1{,}200)(0.3727)}{2} = \$321.99/\text{year}$$

Average annual total cost including purchase cost:

$$\$321.99 + (2)(2{,}400) + (5)(1{,}200) = \$11{,}121.99/\text{year}$$

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# Solution: Comparison

**Individual ordering**

Order 948.68 units/order of A every 0.3953 years

Order 489.90 units/order of B every 0.4083 years

$C_I\ (894.48{,}447.24) = \$11{,}147.75/\text{year}$

**Joint ordering**

Order 894.48 units/order of A and 447.24 Units/order of B every 0.3727 years

$C_J\ (894.48{,}447.24) = \$11{,}121.99/\text{year}$

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# Example: Michaels

| | Amateur | SemiPro | Pro |
|---|---|---|---|
| Annual demand rate | 12,000 | 1,200 | 120 |
| Fixed order cost | $4,000 | $4,000 | $4,000 |
| Product-specific fixed order cost | $1,000 | $1,000 | $1,000 |
| Unit cost | $500 | $500 | $500 |
| Annual inventory carrying rate | 0.2 | 0.2 | 0.2 |

- Michaels sells three types of a product for different market segments.
- Each order incurs a fixed order cost as well as an additional product-specific fixed order cost for each product type included in the order.

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# Solution: Michaels (individual)

| | Amateur | SemiPro | Pro |
|---|---|---|---|
| Annual demand rate | 12,000 | 1,200 | 120 |
| Fixed order cost | $4,000 | $4,000 | $4,000 |
| Product-specific fixed order cost | $1,000 | $1,000 | $1,000 |
| Unit cost | $500 | $500 | $500 |
| Annual inventory carrying rate | 0.2 | 0.2 | 0.2 |

$$q_L^* = \sqrt{\frac{(2)(12{,}000)(4{,}000 + 1{,}000)}{(0.2)(500)}} = 1{,}095.45 \text{ units}$$

$$q_M^* = \sqrt{\frac{(2)(1{,}200)(4{,}000 + 1{,}000)}{(0.2)(500)}} = 346.41 \text{ units}$$

$$C_I(q_L^*, q_M^*, q_H^*) = \$155{,}140$$

$$q_H^* = \sqrt{\frac{(2)(120)(4{,}000 + 1{,}000)}{(0.2)(500)}} = 109.55 \text{ units}$$

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# Solution: Michaels (joint)

| | Amateur | SemiPro | Pro |
|---|---|---|---|
| Annual demand rate | 12,000 | 1,200 | 120 |
| Fixed order cost | $4,000 | $4,000 | $4,000 |
| Product-specific fixed order cost | $1,000 | $1,000 | $1,000 |
| Unit cost | $500 | $500 | $500 |
| Annual inventory carrying rate | 0.2 | 0.2 | 0.2 |

$$T^* = \sqrt{\frac{(2)(4{,}000 + 1{,}000 + 1{,}000 + 1{,}000)}{[(100)(12{,}000) + (100)(1{,}200) + (100)(120)]}} = 0.1025 \text{ years}$$

$$q_L^* = (0.1025)(12{,}000) = 1{,}230.25 \text{ units}$$

$$q_M^* = (0.1025)(1{,}200) = 123.02 \text{ units}$$

$$q_H^* = (0.1025)(120) = 12.30 \text{ units}$$

$$C_J(T^*) = \$136{,}528$$

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# Joint ordering: Epilogue

- Ordering each item individually and ordering all items jointly are two extreme policies.

- There exists other policies where different subsets of items can be ordered together, which is a difficult enumerative problem. Hence, finding the global optimum solution is challenging for multi-item settings.

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# Example: Spring 2022 exam question

Consider an item for which a price discount scheme is in effect. The breakpoints are 3,000 units and 6,000 units. Suppose that you have calculated the economic order quantity values at each price level as follows:

| Unit price ($/unit) | Corresponding EOQ value (units) |
|---|---|
| $c_0$ | 5,282.71 |
| $c_1$ | 5,940.89 |
| $c_2$ | 7,947.20 |

1. Provide the set of candidate solutions for the optimal solution to this problem, if an <u>all-units</u> discount scheme is in effect. Please justify your work briefly.
2. Provide the set of candidate solutions for the optimal solution to this problem, if an <u>incremental</u> discount scheme is in effect. Please justify your work briefly.

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# Case 1: All units

| Unit price ($/unit) | Range | Corresponding EOQ value (units) |
|---|---|---|
| $c_0$ | $0 < q < 3{,}000$ | 5,282.71 |
| $c_1$ | $3{,}000 \leq q < 6{,}000$ | 5,940.89 |
| $c_2$ | $6{,}000 \leq q < \infty$ | 7,947.20 |

Case 1: If $q_j^{EOQ} < b_j$, then $q_j^* = b_j$.

Case 2: If $b_j \leq q_j^{EOQ} < b_{j+1}$, then $q_j^* = q_j^{EOQ}$.

Case 3: If $b_{j+1} \leq q_j^{EOQ}$, then do nothing.

$q_0^{EOQ} = 5{,}282.71$ units; Compare against the range $0 < q < 3{,}000$;
Case 3 above applies (do nothing);
no candidate solution in this range.

$q_1^{EOQ} = 5{,}940.89$ units; Compare against the range $3{,}000 \leq q < 6{,}000$;
Case 2 above applies (between the bounds);
$q_1^* = q_1^{EOQ} = 5{,}940.89$units.

$q_2^{EOQ} = 7{,}947.20$ units; Compare against the range $6{,}000 \leq q < \infty$;
Case 2 above applies (between the bounds);
$q_2^* = q_2^{EOQ} = 7{,}947.20$ units.

Hence, the optimal solution $q^* \in \{5{,}940.89, 7{,}947.20\}$.

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# Case 2: Incremental

| Unit price ($/unit) | Range | Corresponding EOQ value (units) |
|---|---|---|
| $c_0$ | $0 < q < 3{,}000$ | 5,282.71 |
| $c_1$ | $3{,}000 \leq q < 6{,}000$ | 5,940.89 |
| $c_2$ | $6{,}000 \leq q < \infty$ | 7,947.20 |

Case 1: If $q_j^{EOQ} < b_j$ then $q_j^* = b_j$ .

Case 2: If $b_j \leq q_j^{EOQ} \leq b_{j+1}$ then $q_j^* = q_j^{EOQ}$.

Case 3: If $b_{j+1} \leq q_j^{EOQ}$ then $q_j^* = b_{j+1}$ .

$q_0^{EOQ} = 5{,}282.71$ units; Compare against the range $0 < q < 3{,}000$;
Case 3 above applies (larger than the upper bound);
$q_0^* = b_1 = 3{,}000$.

$q_1^{EOQ} = 5{,}940.89$ units; Compare against the range $3{,}000 \leq q < 6{,}000$;
Case 2 above applies (between the bounds);
$q_1^* = q_1^{EOQ} = 5{,}940.89$units.

$q_2^{EOQ} = 7{,}947.20$ units; Compare against the range $6{,}000 \leq q < \infty$;
Case 2 above applies (between the bounds);
$q_2^* = q_2^{EOQ} = 7{,}947.20$ units.

Hence, the optimal solution $q^* \in \{3{,}000, 5{,}940.89, 7{,}947.20\}$.

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